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The higher the better? A study on the dark side of high engagement atmosphere in live streaming sales on consumer post-purchase satisfaction

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ABSTRACT

Research on live streaming engagement has long focused on its positive aspect of promoting consumer conversion but has neglected the potential loss of post-transaction value due to high interaction intensity. This study reveals the "engagement paradox" through a mixed study of actual live streaming and post-purchase evaluation data and survey-based research: although high engagement enhances immediate purchase intention, it significantly reduces post-purchase satisfaction through the dynamic imbalance of cognitive-emotional resources. Based on the integrated framework of elaboration likelihood model, expectation confirmation theory, information overload theory, and flow theory, it is found that the paradox stems from the contextual differentiation of dual paths contextual differentiation, i.e., information dissemination and high interactivity features in live sales contexts trigger information overload and flow experiences, which can lead to satisfaction decay by amplifying the expectation-confirmation gap. This finding bridges the perceived divide in the existing literature on the paradoxical validity of engagement. Further applying gender schema theory, the study breaks new ground by finding that male consumers exhibit more significant flow attrition effects in high-engagement live streaming due to heuristic decision-making tendencies, providing key boundary conditions for gender marketing theory in digital contexts. And these findings also provide practical guidance for designing consumer engagement strategies on live-streaming platforms.

1. Introduction

Live streaming is rapidly growing globally due to its direct and effective communication style and has emerged as a critical factor influencing consumers' purchasing decisions. While the high engagement atmosphere has successfully stimulated consumers' immediate purchase intentions, companies still face a significant challenge in the high return rates observed post-purchase (Xu et al., 2023). This discrepancy between pre- and post-purchase behaviors highlights the need to incorporate the post-purchase phase into studies on consumer engagement. Therefore, understanding and revealing the mechanisms by which the atmosphere of engagement in live streaming affects consumers' post-purchase satisfaction has become a critical issue to be addressed.

Consumer engagement, a hallmark of live-streaming commerce, encompasses three dimensions: cognitive, emotional, and behavioral engagement (Hollebeek and Macky, 2019). A large amount of existing literature focuses on how live streamers' personal characteristics and behavioral performance trigger consumer engagement during live streaming (Busalim et al., 2021; Chen et al., 2022a). For example, the live streamer's speech rate and facial attractiveness (Gao et al., 2023) and language style (Lee and Theokary, 2021) play an important role in positive consumer attitudes. However, less research has focused on consumers' subsequent attitude recognition. Therefore, this study attempts to fill this gap in relevant research by exploring the impact of engagement atmosphere on post-purchase consumer satisfaction.

Therefore, this study proposes three corresponding research questions.

RQ1. How does a highly engaged atmosphere during live streaming affect post-purchase consumer satisfaction?
RQ2. What are the specific mechanisms of influence between the two?
RQ3. What boundary conditions are affected in the mechanism?

We address these questions through complementary studies. The first part employs real-world live-streaming data to uncover the relationship between high engagement atmosphere and post-purchase satisfaction.

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0969-6989/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Consumer engagement is measured using the number of likes, comments and viewers during the live stream, while post-purchase satisfaction is assessed based on the positive rating of exclusive product links associated with each streamer. The results showed that the positive emotions inspired by high engagement failed to carry over to the post-purchase stage and instead reduced consumer satisfaction. Second, we explored the mechanism of action through a questionnaire method, with data from both domestic and foreign samples. The results show that perceived information overload and flow experience mediate the relationship between different engagement atmospheres and post-purchase satisfaction and that gender differences affect this process. This study advances live-streaming research through three contributions. First, integrating expectation-confirmation theory, we bridge pre- and post-purchase phases, revealing how engagement-driven expectations create post-purchase dissatisfaction when unmet—addressing a critical gap in prior work. Second, we challenge the assumed positive link between engagement and purchase intention, demonstrating that high-engagement atmospheres paradoxically reduce satisfaction. Third, a dual-process mechanism explains how information overload (from excessive interactions) and flow experience (via immersive experiences) mediate this effect. Gender moderates these pathways: males exhibit stronger dissatisfaction from flow states. Methodologically, multi-method evidence strengthens validity. Practically, we advise streamers to balance interactivity with product transparency and platforms to implement "cooling-off" nudges during high-engagement sessions. These insights refine engagement strategies to enhance satisfaction and reduce returns, offering actionable guidance for the evolving livestream commerce landscape.

This study is organized as follows: Section 2 provides a literature review, Section 3 develops the research hypotheses, Section 4 elaborates on the methodology, data analysis, and findings, while Section 5 concludes with a discussion of the results and potential avenues for future research.

2. Literature review

2.1. Expectation confirmation theory

Expectation confirmation theory (ECT) was introduced by Oliver in 1980. The theory focuses on consumer satisfaction and covers both pre-purchase and post-purchase phases (Oliver, 1980). Before a purchase, consumers establish expectations, which later serve as a baseline for evaluating the product's actual performance after use, and get the level of confirmation of expectation, which then influences the post-purchase attitudes and repurchase intentions (Halilovic and Cicic, 2013; Oghuma et al., 2016).

While foundational research emphasized static confirmation in traditional e-commerce (Liao et al., 2017), contemporary adaptations integrate social and relational dimensions, particularly in live commerce contexts. The interactive nature of live-streaming commerce transforms ECT dynamics through real-time social engagement, where immersive interactions with sellers create " parasocial interaction " that recalibrate expectation thresholds (Joo and Yang, 2023; Wang et al., 2024). Unlike conventional transactional e-commerce's product-centric static confirmation processes, live streaming environments introduce platform-mediated social cues (e.g., real-time communication, parasocial interactions) that simultaneously influence both expectation dynamics and post-purchase evaluation (Yang et al., 2023). This study extends ECT by hypothesizing that varying engagement intensities during live streams differentially modulate confirmation mechanisms and satisfaction outcomes (Kumar and Anjaly, 2017; Chen et al., 2018; Isaac and Marks, 1994; Churchill and Carol, 1982). While ECT is widely recognized in online shopping research, the specific interplay between different live-streaming engagement atmospheres and post-purchase satisfaction across different stages remains unexplored.

2.2. Elaboration likelihood model

Live streaming commerce's experiential nature stimulates unplanned purchases through dual information processing pathways (Xiang et al., 2016). This study employs the elaboration likelihood model (ELM) to examine how consumers cognitively engage with complex content combining argument quality and heuristic cues (Luo et al., 2024b). To explore consumers' information processing during live shopping, this study introduces ELM theory. ELM posits two persuasion routes: the central path involving systematic evaluation of message content (e.g., product specifications), and the peripheral path relying on contextual cues like streamer credibility (Petty et al., 1981; Gao et al., 2021).

Unlike traditional e-commerce's central-path dominance, live commerce synergizes both pathways through real-time interactions and multimedia immersion (Sun et al., 2019). Streamers strategically deploy central cues (expert demonstrations) and peripheral cues (audience enthusiasm) to optimize persuasion (Geng et al., 2021; Guo et al., 2022). The platform's dynamic environment amplifies cognitive-affective duality: detailed product analysis coexists with heuristic processing of social proof and emotional contagion. These mechanisms align with ECT, where engagement-driven expectations evolve into post-purchase evaluations through ELM-mediated interactions. By integrating ELM and ECT frameworks, this research elucidates the unique cognitive-affective interplay in live streaming commerce, offering a novel perspective on how digital immediacy transforms consumer decision-making processes.

2.3. Engagement in live-streaming

Consumer engagement has emerged as a critical concept in marketing literature, recognized for its multidimensional nature encompassing cognitive, emotional, and behavioral dimensions (Hollebeek et al., 2014; Hollebeek and Macky, 2019). This construct extends beyond transactional interactions (van Doorn et al., 2010) to influence customer loyalty (Rather, 2018; Thakur, 2019), industrial innovation (Hollebeek et al., 2022), and service value co-creation (Viglia et al., 2023). The interactive nature of live streaming platforms amplifies engagement's complexity, where instant feedback and personalized communication with streamers exemplify its multidimensional character (Kang et al., 2021). In digital contexts like live streaming, engagement manifests through real-time interactions including viewing, commenting, and liking(Barger et al., 2016), while also involving emotional responses to content and cognitive processing of brand information. Empirical evidence confirms significant correlations between these engagement behaviors and purchase intentions (Zheng et al., 2022), mirroring patterns observed in social media engagement research (Alhabash et al., 2015; Sharon and Morgan, 2012). This study specifically examines the holistic engagement atmosphere within live-streaming environments, recognizing its pivotal role in shaping contemporary consumer behavior. By integrating established theoretical frameworks with platform-specific dynamics, the research aims to advance the understanding of engagement mechanisms in emerging digital marketing contexts.

2.4. Gender schema theory

Gender schema theory (GST) posits that gender-based cognitive frameworks shape differential information processing and behavioral patterns between males and females (Sandra Lipsitz, 1981). Empirical evidence reveals systematic gender divergences: males typically exhibit selective cognitive processing and achievement-oriented motivations, whereas females demonstrate comprehensive information integration driven by relational needs (McClelland, 1975; Meyers-Levy, 1991; Smith and Leaper, 2006). These cognitive differences manifest in distinct social orientations - females tend toward interdependent relationship-building, while males emphasize independent

self-conceptualization (Eagly and Wood, 1991). Recent applications in digital commerce contexts reveal gender-specific behavioral patterns, with females demonstrating heightened sensitivity to hedonic value in impulse purchases (Hashmi et al., 2019). Similar gendered disparities persist across technology adoption studies, including social media engagement (Lai et al., 2019), smartphone usage (Ameen et al., 2020) and mobile app usage (Douglas, 2019). Despite established gender differences in conventional digital environments, the interactive dynamics of live-streaming commerce remain underexplored. This research gap is particularly significant given live streaming's unique combination of real-time social interaction and commercial persuasion, which may amplify or modify traditional gender-based consumption patterns.

Table 1 summarizes some of the research highlights on engagement in live streaming.

3. Research hypothesis

3.1. High engagement atmosphere and post-purchase satisfaction

In prior literature, engagement in live streaming is typically measured by metrics such as likes, comments, and shares (Kang et al., 2021). Studies suggest that consumer engagement in live-streaming significantly shapes their attitudes: Shen et al. (2022) drawing on parasocial interaction theory, argued that consumers develop positive purchase intentions due to parasocial connections, which foster emotional involvement. Similarly, Luo et al. (2024b), using the ELM theory, suggested that increasing consumer engagement during live streaming enhances the likelihood of impulse purchases. It follows that in real-time live-streaming contexts, consumers are easily drawn into engagement and can be influenced by the highly interactive environment (Feng et al., 2024).

Expectation confirmation theory spans both the pre - and post-purchase phases of consumer behavior. Before making a purchase, consumers form initial expectations based on their preliminary understanding of the product (Oliveira et al., 2017). After the purchase, they evaluate the product by comparing it to these expectations, determining whether the expectations have been met or exceeded. This post-purchase evaluation influences overall satisfaction (Wang et al., 2016). Higher levels of satisfaction tend to occur when the confirmation exceeds initial expectations (Ma et al., 2023). In live-streaming environments, previous research has highlighted the significant impact of high-arousal live streamers on consumers' impulse purchasing behavior (Luo et al., 2024b). Customer engagement not only enhances the intimacy with live streamers (Luo et al., 2024b) but also reduces customer uncertainty and consumption concerns and enhances impulse purchase intentions (Lo et al., 2022). This evidence suggests that high engagement atmosphere within live-streaming can lead to inflated consumer expectations (Wang et al., 2024), which may negatively affect the perceived quality of service and overall experience post-purchase (Feng et al., 2024). Thus, from the perspectives of consumer-to-consumer and consumer-live streamer interactions, we hypothesize that high levels of engagement during live streaming will adversely impact post-purchase satisfaction.

H1. High engagement atmosphere in live-streaming environments negatively affects post-purchase satisfaction.

Table 1
Relevant studies on engagement.

	Authors	Research Topic
Influences on the atmosphere of engagement in the live streaming room	Kang et al. (2021)	Interactivity
	Wongkitrungrueng and Assarut (2020)	Perceived value
	Hilvert-Bruce et al. (2018)	Social motivations
	Tian and Frank (2024)	Live streaming features
	Luo et al. (2024a)	Streamers' topic types
	Yan et al. (2023)	IT affordances
	(Chen and Yin, 2024)	Streamer characteristics and cultural traits
	Diwanji et al. (2020)	Copresence
Aftereffects on the atmosphere of engagement in the live streaming room	Zhang and Xu (2024)	Value co-creation
	Luo et al. (2024b)	Impulsive buying tendency
	Shen et al. (2022)	Purchase intention
	Zhang et al. (2023)	High-value gifting
	Liu et al. (2025)	Value for live streaming platforms
	Fan et al. (2024)	Live-streaming sales mode
	Chen et al. (2022a)	Product sales
Our study	/	Post-purchase satisfaction

3.2. High engagement atmosphere with perceived information overload and flow experience

User engagement has demonstrated cross-disciplinary applicability. In gamified environments, immersive elements such as challenges, rewards, and social interaction mechanisms significantly enhance participation and shape user experiences (Harwood and Garry, 2015; Hofacker et al., 2016; Xi and Juho, 2020). Social commerce extends these social-driven mechanisms by integrating community-based consumption patterns (Li, 2019). Empirical evidence suggests that social influence surpasses information quality in shaping purchase intentions, though excessive interaction may induce social fatigue (Fu et al., 2020; Liu et al., 2021). Live-stream commerce represents a novel convergence, merging gamified immersion with social commerce interactivity. This integration transforms consumer decision-making processes, transitioning from static information processing to dynamic experiential engagement. The hybrid model leverages real-time interactivity and entertainment elements to optimize participatory consumption experiences while navigating potential engagement paradoxes.

Live streaming continues to experience rapid growth (Huang et al., 2024), with consumer engagement emerging as a key feature of the medium (Chen et al., 2022b). Consumer engagement in live-streaming contexts is frequently defined as a multidimensional construct encompassing cognitive, emotional, and behavioral aspects (Hollebeek et al., 2023). This is exemplified by actions such as liking, commenting (e.g., Danmaku), and sharing (Kang et al., 2021). These interactions also generate various forms of information that audiences must process during live streaming sessions. Consumers approach live streaming sessions with pre-established rational expectations about product attributes (Huang et al., 2024). In the initial low-interactivity phase, cognitive processing predominantly follows the central route, with viewers systematically analyzing live streamer-delivered product specifications (Chen et al., 2022a). As participatory intensity escalates through heightened host-viewer dialogues and viewer-to-viewer social dynamics (Yen et al., 2024), peripheral cues triggered by hedonic stimuli gradually dominate decision-making processes. This cognitive transition fosters flow states that enhance session immersion (Xiang et al., 2016; Sarwar et al., 2024), yet concurrently elevates aspiration levels through emotional contagion (Wang et al., 2024), ultimately compromising objective information evaluation. The persistence of central route processing manifests as a countervailing mechanism, wherein consumers attempt to reconcile affective engagement with analytical scrutiny to reduce anticipated post-purchase dissonance (Jia et al., 2024). However, this cognitive duality induces three synergistic impairments: attentional tunneling that narrows critical attribute assessment (Wu et al., 2020), cognitive overload disrupting working memory consolidation (Shen et al., 2022), and decision paralysis arising from conflicting neural pathway outputs. These compounded effects systematically degrade decision-making quality through neural resource

depletion, despite consumers' conscious efforts to maintain rational appraisal.

In the AI era, algorithmic prediction engines enable hyper-personalized curation through interest-mapping neural networks (Lv et al., 2024). VR/AR systems amplify multisensory immersion via emotionally charged spatial computing (Huang and Chung, 2024), while real-time reciprocity norms cultivate urgency-driven impulse consumption (Li et al., 2024). This highly immersive environment exacerbates information overload and the flow experience, making it difficult for consumers to rationally weigh information. Specifically, the wide variety of forms of information (including likes, comments, and shares) posted in the live streaming room, while attracting brand-aware consumers through the richness of the information shared (Zhao et al., 2023). However, the abundance of information can create challenges in processing, leading to information overload. This occurs when the amount of information exceeds an individual's capacity to process it, which can significantly impact the generation of impulse purchases (Zhang and Ahmad, 2024). Based on this understanding, we posit that various forms of consumer engagement increase the breadth of information, potentially complicating content evaluation in an environment characterized by information overload (Gottschalk et al., 2020).

In addition, the engaging real-time chat feature in live streaming enables instant feedback loops (Mohr and Sohi, 1995). Against the backdrop of high arousal from streamers, the influx of likes, comments, and shares makes consumers perceive a strong sense of interactivity (Joo and Yang, 2023). This interactivity can give audiences a sense of spatial immediacy, making them feel like they are in the same place at the same time as the streamer and other audiences (Xu et al., 2023). Flow experience constitutes a critical psychological mechanism in e-commerce and live-streaming contexts. Flow transcends user-content interaction to encompass cognitive absorption and perceptual immersion (Hoffman and Novak, 1996). Based on flow theory, Wu et al. (2016) verified that a high degree of interactivity in the context of live streaming has a positive impact on the flow experience. Li and Peng (2021) revealed the truth that live streaming scenario characteristics (telepresence and entertainment) can stimulate users' flow experience in live streaming. As a result, the highly interactive and immediate feedback mechanisms in live streaming environments can attract consumers to engage more deeply and significantly enhance the flow experience (Sun et al., 2019; Yim et al., 2017).

Therefore, we propose the following hypothesis.

H2. High engagement atmosphere positively influences on consumers' perceived information overload

H3. High engagement atmosphere positively affects consumers' flow experience

3.3. Multiple mediating effects of perceived information overload and flow experience from the ELM perspective

While live streaming offers a diverse and real-time flow of information, the problem of information overload cannot be ignored. According to Jacoby (1984), the volume of information and the time available for decision-making significantly influence outcomes. Information overload theory suggests an inverted U-shaped relationship: initially, decision quality improves with more information, but beyond a certain point, additional information leads to poorer decisions. For instance, Furner et al. (2016) found that moderate levels of information load are most effective in fostering high trust and purchase intentions. Factors such as information complexity (Paul and Nazareth, 2010), information amount (Gao et al., 2018), and time constraints (Scheibehenne et al., 2010) are critical contributors to information overload. Consequences of information overload include diminished user satisfaction (Liang et al., 2006) and increased negative emotions (Swar et al., 2017). In retail contexts, information overload has been explored in areas such as in-store mobile usage during omnichannel shopping (Fang et al., 2021) and consumer decision-making in online shopping (Chen et al., 2009). In highly interactive live-stream shopping environments, consumers face information overload when message volume exceeds their processing capacity. This phenomenon directly stems from the intense interactivity of consumption scenarios (Hu and Krishen, 2019; Lee and Theokary, 2021). Building on the ELM, while high-involvement contexts may activate systematic information processing (Cacioppo et al., 1986), excessive stimuli create tension between engagement and cognitive strain, escalating cognitive load (Leppink et al., 2014). When cognitive demands surpass individual thresholds, decision-making difficulties intensify, leading to biased product evaluations and heightened expectations (Wang et al., 2024). The subsequent discrepancy between expectations and actual experiences diminishes post-purchase satisfaction (Malhotra Naresh, 1982), positioning cognitive load as the critical mediator between information overload and satisfaction reduction. Despite its significance, limited research explores this mechanism in live-stream commerce.

Engagement in live streaming often triggers a flow experience, which can drive consumers' impulse purchase intentions (Huang, 2016). Flow is characterized by a highly immersive and enjoyable shopping experience (Joo and Yang, 2023), where individuals become deeply absorbed in the activity at hand (Csikszentmihalyi, 2000). Flow theory provides a valuable framework for understanding the negative consequences of a high-engagement atmosphere on post-purchase consumer satisfaction in live-streaming contexts. Unlike traditional online shopping, live streaming allows consumers greater freedom and control over the shopping experience, including interactions with streamers and other viewers. This enhanced control can improve the overall consumption experience and foster impulsive purchasing behaviors. And in live-stream shopping, consumers reciprocally influence and are influenced by peers' experiences, fostering shared perceptions through heuristic cues (e.g., real-time danmaku comments (Hu et al., 2017)) that activate peripheral route processing. These rapid, emotion-laden interactions amplify affective engagement and impulsive buying (Zeng et al., 2023). Although impulse buying is not inherently negative (Hausman, 2000), it is often linked to adverse outcomes, such as financial strain, reduced self-esteem, and post-purchase dissatisfaction (Rook, 1987).

Although information overload and the flow experience are distinct attitudinal perceptions, they can coexist in a highly engaging live-streaming environment, as informed by ELM theory. Both constructs may contribute to low-quality decision-making, ultimately resulting in decreased post-purchase satisfaction. Therefore, we propose the following hypothesis.

H4. Consumers' post-purchase satisfaction is adversely impacted by perceived information overload.

H5. Flow experience negatively affects consumer post-purchase satisfaction.

Accordingly, we propose.

H6. Perceived information overload plays a mediating role between high engagement atmosphere and low post-purchase satisfaction.

H7. Flow experience mediates high engagement atmosphere and low post-purchase satisfaction.

3.4. Moderating role of gender

Gender differences significantly affect the impact of marketing strategies on consumer behavior (Puccinelli et al., 2013). Numerous studies have shown that there are significant differences between genders in terms of information processing styles and shopping preferences. Women tend to adopt a more holistic information processing strategy, incorporating emotional and subjective factors into their judgments, whereas men typically focus on objective attributes through selective

processing (Meyers-Levy, 1988; Meyers-Levy and Durairaj, 1991). Research indicates that women are more likely to engage in imagination and mental simulation while shopping (Isaac and Marks David, 1994) and respond more to the shopping experience, while men prioritize functional features (Dittmar et al., 2004). Furthermore, women demonstrate higher involvement in information searching (Kim et al., 2007), whereas men prefer quicker decision-making (Ganguly et al., 2010). In terms of purchasing behavior, women generally shop at a more leisurely pace, focusing on long-term needs and hedonic experiences (Kim et al., 2007), while men are more attentive to value recognition (Chia-Chen et al., 2018). Consequently, the cognitive and affective differences brought about by this gender difference may affect consumers' experience and decision-making in online shopping (Tsichla et al., 2016). This difference is also coming to the fore in the live-stream shopping ecosystems. Persuasive cue processing diverges significantly, with females demonstrating heightened substitutive product adoption under high-source-credibility conditions (Peng et al., 2024). Visual information density modulates affective responses differentially - males exhibit elevated arousal in complex interfaces, while females show purchase intent peaking at moderate stimulation levels. Trust formation mechanics follow distinct neural pathways: product expertise framing predominantly enhances male source credibility, whereas longitudinal para-social exposure disproportionately strengthens female trust (Xiong et al., 2024).

In a high-engagement live streaming environment, we anticipate that more emotional women will be more inclined to consider information holistically and experience flow, potentially leading to higher perceived product value. Conversely, due to their selective processing of information, relatively more rational men may be more sensitive to perceptions of information overload and flow experiences, resulting in lower post-purchase satisfaction evaluations.

Building on this literature, we propose the following hypotheses.

H8. Gender moderates the relationship between perceived information overload and post-purchase satisfaction in different engagement atmospheres.

H9. Gender moderates the relationship between flow experience and post-purchase satisfaction in different engagement atmospheres. Fig. 1 present the theoretical model diagram.

4. Method

4.1. Study1

4.1.1. Douyin

Study 1 aimed to investigate the relationship between different levels of engagement atmosphere and consumer post-purchase satisfaction in an actual online environment. We sourced live streaming data from the popular social networking platform Douyin, utilizing Feigua Douyin, a third-party data analytics platform. This approach has been employed in previous studies (Ren et al., 2023; Peng et al., 2024; Xu et al., 2024). Feigua Douyin collects public data—including metrics such as fan counts, followers, comments, retweets, and likes—via a crawler program and provides real-time monitoring and organization of live streaming data. To ensure data reliability, three graduate students were enlisted to verify the data. Post-purchase satisfaction was assessed by analyzing the rate of positive reviews (number of positive reviews/total number of reviews) associated with products through exclusive links (Jones and Taylor, 2018), with this metric selected to control for potential confounding effects from external review sources. Consumer engagement was quantified by counting the number of likes, comments and views received (Zheng et al., 2022), numerous studies have validated their effectiveness across different product types and live streaming models (Clement et al., 2021; Zheng et al., 2022; Luo et al., 2024a).

The sample comprised 235 product explanation clips from live streaming. The third-party platform updates daily rankings of live-streaming talent and associated live-streaming data, including metrics such as real-time likes, comments, views, and product explanation duration. To ensure consistency in consumer engagement measurement, we selected products that were featured for more than 1 min and only once during the live stream (Lin et al., 2021). Given the time lag

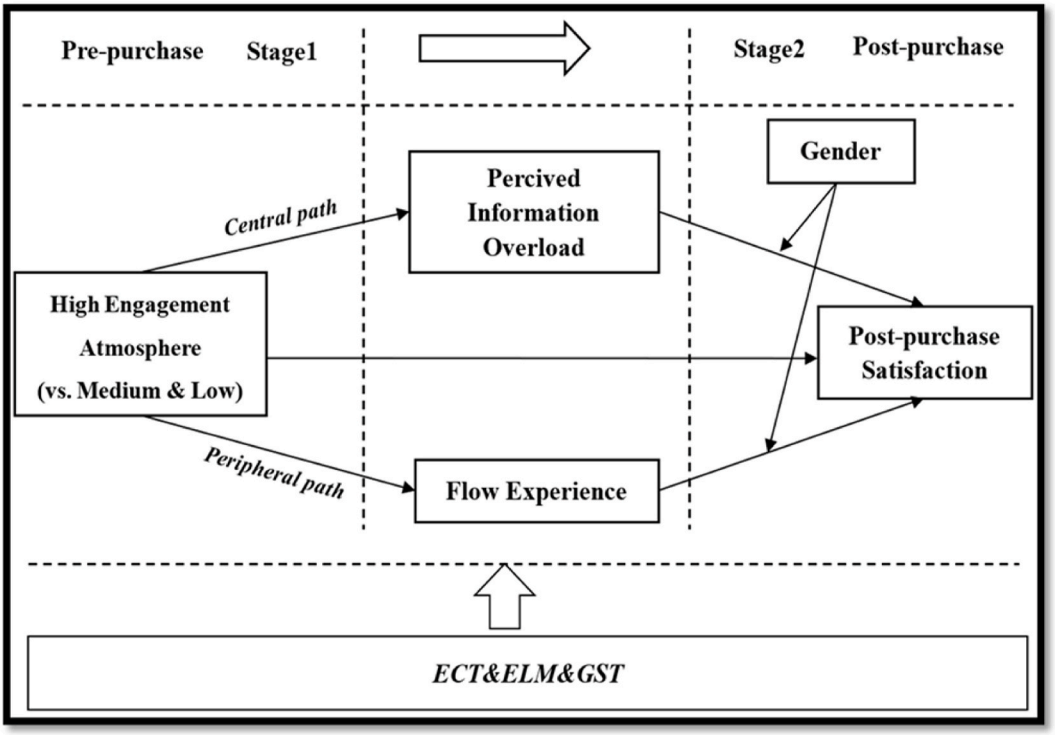


Fig. 1. Theoretical model.

between purchase and post-purchase reviews, we focused on live-streaming data from talent rankings between July 20 and July 22, 2024, and processed the data from August 1 to August 3, 2024, which is not affected by major promotions. All values of the live streaming data provided by this data source are presented in their entirety and there are no missing values. Across all product explanations, a total of 20,468,742 likes were recorded (M = 146205, SD = 290018.960, Skewness = 2.996). There were 511,203 comments (M = 3651, SD = 6071.100, Skewness = 2.770). Additionally, the average number of online viewers across all product presentations totaled 67775667 (M = 28841, SD = 30355.490, Skewness=2.221).

In this section, we employ an independent-sample *t*-test to evaluate whether high engagement atmosphere—characterized by increased likes, comments and views in the live streaming environment—is negatively correlated with consumer post-purchase satisfaction (see Fig. 2 and Fig. 3). To differentiate between high, medium, and low levels of consumer engagement, we adopt the classification method established by Calvo-Porral et al. (2021). We first calculate the engagement score for each consumer, sum these scores, and then compute the average. Following the methodology of (Quester and Smart, 1998), we employ a tertile segmentation approach to categorize consumers based on their engagement levels. Individuals in the lowest quartile are classified as low engagement, those in the middle quartile as medium engagement, and participants in the upper quartile as high engagement atmosphere (Zaichkowsky, 1985). Engagement levels are denoted as 0 for low, 1 for medium, and 2 for high, effectively creating three distinct groups. $M_{LoM\ w}$ = 9545.873, SD = 2676.948; M_{Medium} = 36046.449, SD = 29058.896; M_{High} = 426202.653, SD = 312984.236. A one-way ANOVA reveals significant differences in post-purchase satisfaction across engagement levels ($\boldsymbol{F = 7.117,\ p = 0.001}$). Bonferroni post-hoc comparisons show that post-purchase satisfaction for high engagement atmosphere is significantly lower than for low engagement. All analytical procedures were preceded by verification of normal distribution assumptions and homogeneity of variance. Complete statistical outputs are documented in Appendix B. Thus, the findings suggest that increased engagement levels may lead to lower post-purchase satisfaction, indicating that high engagement atmosphere during live broadcasts may have a potentially negative impact on consumer satisfaction. Fig. 4 and Table 2 present the specific results.

4.1.2. Kuaishou

Following standardized collection protocols mirroring Douyin's methodology, we conducted structured analysis of Kuaishou livestream data encompassing 111 product demonstration segments (see Fig. 5 and 6). The dataset comprised top-ranked broadcasts from January 3-7, 2025, with subsequent data scrubbing occurring January 18-31. From the initial 111 observations, 14 entries with missing viewer interaction metrics were excluded (<12.7 % attrition). In the end, the sample consisted of 97 product explanation clips. Across all product explanations, a total of 1355670000 likes were recorded (M = 13975979.380, SD = 82235120.830,Skewness = 9.312). There were 1298972 comments (M = 13391.464,SD = 8176.519,Skewness = 1.399). Additionally, the average number of online views across all product presentations totaled 1728470.333 (M = 18004.899, SD = 30769.321, Skewness = 3.545). Engagement was similarly categorized into three groups according to the methodology in 4.1.1, M_{Low} = 1685.684, SD = 1011.054; M_{Medium} = 30291.835, SD = 30146.659; M_{High} = 1716656.955, SD = 5087953.973.

A one-way ANOVA was subsequently conducted following confirmation of parametric assumptions (normality and homogeneity of variance) through preliminary testing, detailed in Appendix B. The analysis revealed significant differential impacts of high engagement on consumer post-purchase satisfaction (F=3.731, p=0.028). Bonferroni-adjusted pairwise comparisons demonstrated that high-engagement environments yielded significantly lower satisfaction than low-engagement conditions. This cross-platform pattern suggests a consistent negative association between elevated live-stream interaction levels and post-consumption evaluation outcomes. Fig. 7 and Table 3 present the specific results.

Our analysis of live-streaming data from Douyin and Kuaishou reveals compelling empirical evidence for the engagement atmospherics paradox: High engagement during live streaming negatively impacts post-purchase satisfaction, which strongly support H1. And the above dataset is very robust, with 235 product explanation clips included in the data from the Douyin platform. These products encompass different sectors of the market - from milk to skincare. The data from the Kuaishou platform contains 97 product explainer clips. These products contain different sectors of the market - from seafood products to color cosmetics.

In the current study, marketing managers are looking to increase

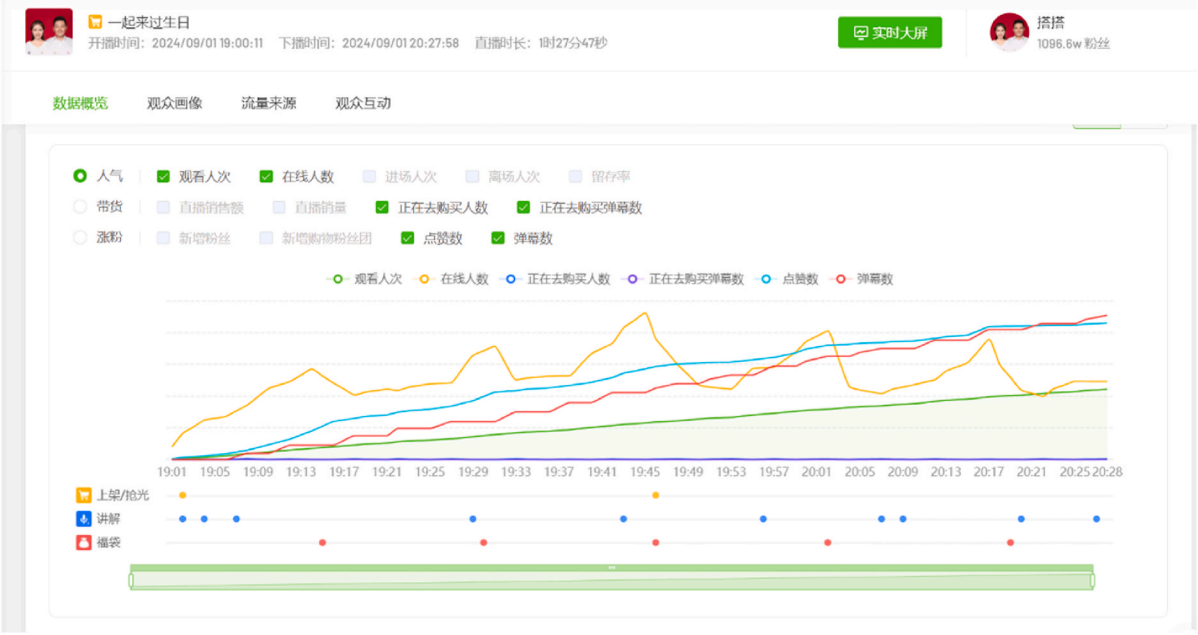


Fig. 2. Live streaming data presentation - Douyin.



Fig. 3. Information about specific products - Douyin.

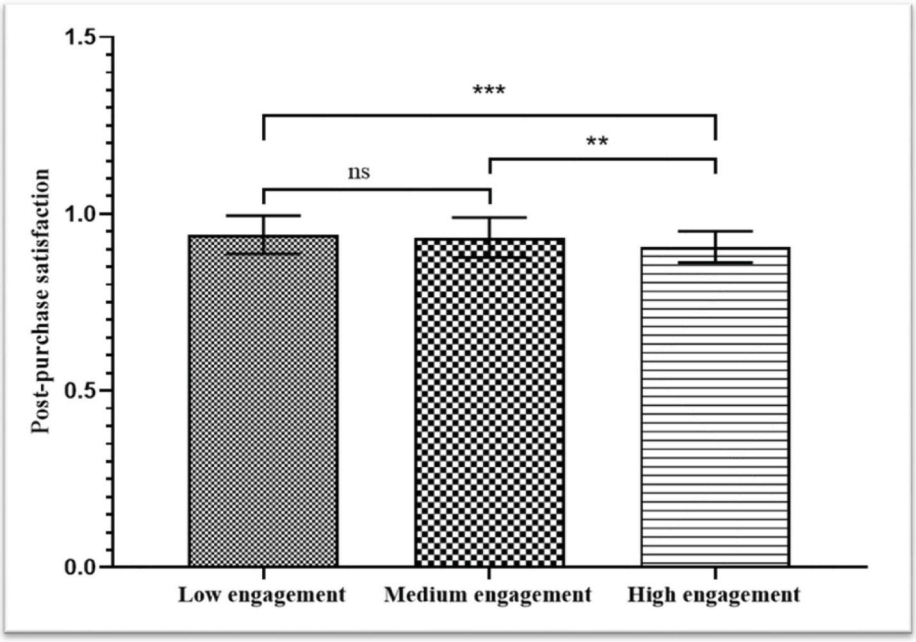


Fig. 4. Differences in the impact of engagement on post-purchase satisfaction.

Table 2
Impact of different levels of engagement atmosphere on consumer post-purchase satisfaction.

Groups	Post-purchase satisfaction	F- test		Multiple Means Comparison
	$\overline{X} \pm S$	F	P	
Low engagement	0.941 ± 0.054	7.117	0.001	Low engagement > High engagement; Medium engagement > High engagement
Medium engagement	0.933 ± 0.056			
High engagement	0.907 ± 0.044			

consumer engagement in the live-streaming process as a sign of good live-streaming (Zhai and Chen, 2023). Also in this sample, 84 segments were high-engagement livestreams, accounting for nearly 30 %. This fully shows that the managers have not yet found the negative impact of high engagement atmosphere.

However, despite this preliminary finding, the current research still leaves some important problems. Due to the use of live streaming data, the study cannot determine whether there is a causal relationship between high engagement atmosphere and consumer satisfaction after purchase. In addition, in this study, actual consumer behavior, such as likes and comments, was used to measure consumer engagement, and positive reviews were used to measure consumer satisfaction. However, marketers may question whether there is a similar impact between this data as a measurement metric and traditional marketing metrics. In addition, marketers may question the process by which high engagement atmosphere influences subsequent consumer outcomes, and identifying the processes behind observed effects is also important to identify the mechanisms of impact of the effects and their application in new contexts. To address these issues, we need to conduct surveys to assess their direct and indirect impact on the behavior of the consumers involved. The reason for choosing the questionnaire method is that we can coherently measure the in-process and post-purchase stages of consumers by asking them to recall their live shopping experience, which is also a common method in research (Sergio et al., 2022; Xin et al., 2024).

4.2. Study2

4.2.1. Data collection

To ensure the content validity of the scale, the measured variables in this paper include high engagement atmosphere, perceived information overload, flow experience, and post-purchase satisfaction. Among them, the high engagement measure refers to the high engagement measure developed by (Zhai and Chen, 2023); The scale for perceived information overload is based on the items developed by (Malhotra Naresh, 1984); The flow experience measures refer to the scale developed by (Koufaris, 2002); The post-purchase satisfaction scale refers to the items identified by (Kumar and Bhawna, 2017). All items were measured on a 7-point Likert scale.

We implemented a probability-based sampling procedure utilizing the computerized random number generator embedded in the WJX platform (Liu et al., 2023) to ensure equitable participant selection. Eligibility criteria required verified live-stream purchase experience,

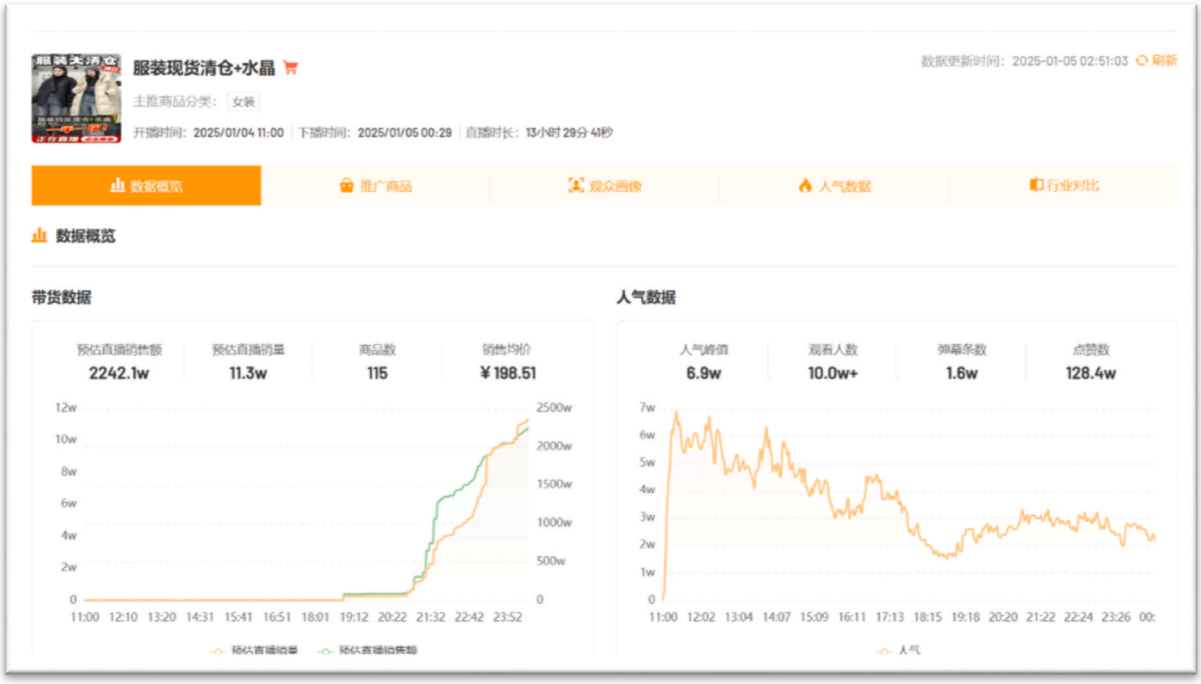


Fig. 5. Live streaming data presentation – Kuaishou.



Fig. 6. Information about specific products– Kuaishou.

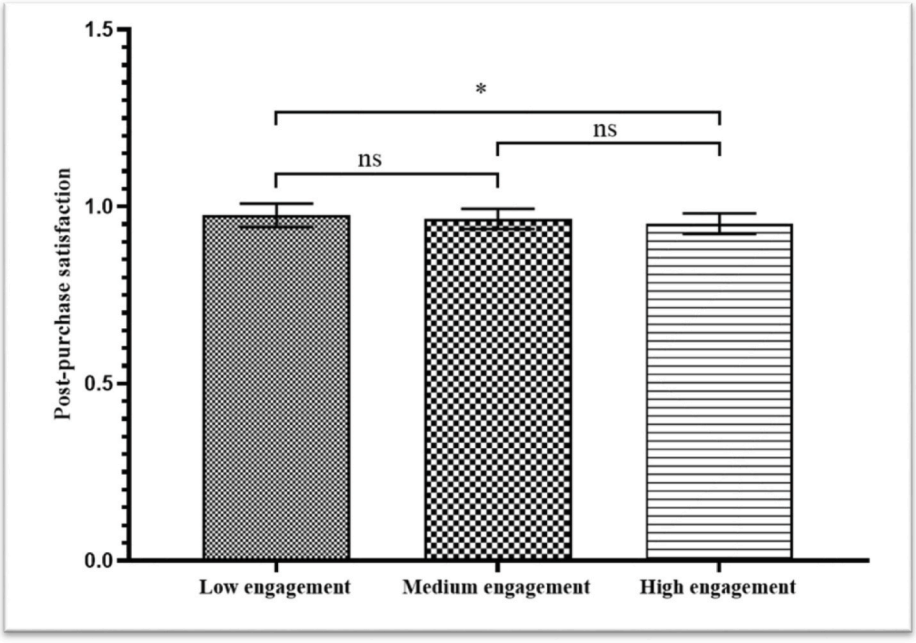


Fig. 7. Differences in the impact of engagement on post-purchase satisfaction.

Table 3
Impact of different levels of engagement atmosphere on consumer post-purchase satisfaction.

Groups	Post-purchase satisfaction	F - test		Multiple means comparison
	$\overline{X} \pm S$	*F*	*P*	
Low engagement	0.975 ± 0.007	3.731	0.028	Low engagement < High engagement
Medium engagement	0.965 ± 0.004			
High engagement	0.952 ± 0.006			

with episodic memory activation through priming questions (e.g., "Please recall your most recent experience of purchasing a product in the live-streaming room and the price of the product you purchased"). A total of 472 responses were collected. After screening for invalid responses—defined as those with excessive repetition or insufficient completion time—we retained 402 valid responses, yielding an effective response rate of 85.17 %. In addition, thanks to the rationality and rigor of the questionnaire setup in this study, the data collected during the questionnaire survey were complete without any missing values. Among the valid questionnaires, the sample comprised 49 % male and 51 % female respondents. The group of 18-40 years old accounted for 73.24 %. This effectively fits the existing characteristics of the mainstream group of live shopping (Gu et al., 2023). The IP addresses of the subjects showed that they came from 19 different urban areas, which reflects the diversity and representativeness of the sample. Table 4 provides detailed statistics.

4.2.2. Reliability and validity analysis

4.2.2.1. Reliability test. The reliability of the sample data was assessed using SPSS 26.0, with the results detailed in Table 5. Cronbach's α values for each variable ranged from 0.814 to 0.856, this indicates that the data from the scale is reliable. Collinearity diagnostics revealed acceptable VIF ranges (1.148-1.249), significantly below the 5.0 threshold. Durbin-Watson analysis confirmed residual independence (DW=1.935; 1.93-1.95=non-autocorrelated range).

Table 4
Demographic characteristics of sample.

Classification	Features	Number	Proportion
Gender	Man	197	49.00 %
	Woman	205	51.00 %
Age	Under 18 years old	17	4.23 %
	18–25	48	11.94 %
	26–30	118	29.35 %
	31–40	127	31.59 %
	41–50	61	15.17 %
	51–60	18	4.48 %
	Above 60 years old	13	3.23 %
Occupation	Students	38	9.45 %
	State-owned Enterprises	92	22.89 %
	Institutions	30	7.46 %
	Civil Servants	19	4.73 %
	Private enterprises	198	49.25 %
	Foreign Invested Enterprises	25	6.22 %
Frequency of viewing live sales	Never	0	0.00 %
	1-2 times per month	93	23.13 %
	1-2 times per week	140	34.83 %
	3-5 times per week	131	32.59 %
	Daily	38	9.45 %
The average time spent watching live streaming	Less than 30 min	48	11.94 %
	30 min - 1 h	198	49.25 %
	1 h - 2 h	116	28.86 %
	More than 2 h	40	9.95 %

Table 5
Reliability and validity analysis of variables.

Variables	Measurements	Factor loading	AVE	CR	Cronbach's α
Perceived information overload (PIO)	PIO 1	0.842	0.698	0.874	0.828
	PIO 2	0.825			
	PIO 3	0.839			
Flow experience (FL)	FL 1	0.818	0.677	0.863	0.814
	FL 2	0.777			
	FL 3	0.871			
Post-purchase satisfaction (PS)	PS 1	0.814	0.666	0.889	0.856
	PS 2	0.814			
	PS 3	0.806			
	PS 4	0.831			

4.2.2.2. Validity test. Average variance extracted (AVE) values for all variables ranged from 0.666 to 0.698, and composite reliability (CR) values ranged from 0.863 to 0.889. These results indicate that the measurement indicators had strong explanatory power for each variable. Furthermore, Table 6 presents the discriminant validity indices, showing significant correlations among the four variables ($p < 0.01$). The coefficient values were lower than the square roots of the AVE, indicating satisfactory discriminant validity between the variables.

4.2.2.3. Common method bias. To assess common method bias, Harman's single-factor test was applied. Results showed that the largest principal component accounted for 27.970 % of the total variance, well below the 40 % threshold. This confirms the absence of significant methodological factors influencing the variance in this study. In addition, we tested for common method bias using single-factor confirmatory factor analysis (CFA). The result of the CFA test shows that x2/df = 1.849, RMSEA = 0.046, GFI = 0.965, NFI = 0.958, IFI = 0.980, TLI = 0.971, CFI = 0.980, while the single-factor CFA's fit index results are x2/df = 21.318, RMSEA = 0.225, NFI = 0.575, IFI = 0.586, TLI = 0.465, CFI = 0.584, the fit index of one-factor CFA is much poorer than the original model, which indicates that the validity of this model passes the test.

4.2.3. Main effect test

Following the methodology described in Section 4.1.1, engagement levels were divided into three categories: high, medium, and low. $M_{Low} = 2.578$, SD = 0.444; $M_{Medium} = 4.285$, SD = 0.659; $M_{High} = 6.290$, SD = 0.507. A one-way ANOVA was conducted to analyze the main effects, revealing how different engagement levels impact post-purchase satisfaction. Normal distribution and variance chi-square tests were rigorously verified prior to analytical procedures. Full diagnostic metrics are tabulated in Appendix B.

The results indicated that the engagement atmosphere had a significant impact on consumers' perceived information overload, flow experience, and post-purchase satisfaction (perceived information overload: $\eta^2 = 0.101$, $F(1, 402) = 22.365$, $p < 0.001$; flow experience: $\eta^2 = 0.119$, $F(1, 402) = 27.007$, $p < 0.001$; post-purchase satisfaction: $\eta^2 = 0.124$, $F(1, 402) = 28.278$, $p < 0.001$). Specifically, the high engagement atmosphere resulted in significantly higher levels of perceived information overload and flow experience compared to medium and low engagement atmospheres (perceived information overload: $M_{High} = 5.092$, $M_{Medium} = 4.217$, $M_{Low} = 3.855$, $p < 0.001$; flow

Table 6
Discriminant validity test of variables.

	1	2	3
Perceived information overload (PIO)	***0.835***		
Flow experience (FL)	0.402**	***0.823***	
Post-purchase satisfaction (PS)	−0.322**	−0.353**	***0.816***

Note: ** means that at the 0.01 level (two-tailed), the correlation is significant.
Note:*p < 0.05, **p < 0.01, ***p < 0.001.

experience: $M_{High} = 5.175$, $M_{Medium} = 4.120$, $M_{Low} = 4.023$, $p < 0.001$). Conversely, the high engagement atmosphere was associated with significantly lower post-purchase satisfaction compared to medium and low engagement atmospheres ($M_{High} = 2.782$, $M_{Medium} = 3.784$, $M_{Low} = 4.012$, $p < 0.001$). Consequently, hypotheses H1, H2, and H3 were supported. Figs. 8, 9 and 10 present the detailed results.

Furthermore, the effects of perceived information overload and flow experience on consumer post-purchase satisfaction were found to be negatively significant (perceived information overload: B = −0.303, SE = 0.045, t = −6.801, p < 0.001, LLCI = −0.390, ULCI = −0.215; flow experience: B = −0.349, SE = 0.046, t = −7.546, p < 0.001, LLCI = −0.440, ULCI = −0.258), thereby supporting hypotheses H4 and H5.

4.2.4. Mediated effects test

To further explore the psychological mechanisms through which these engagement levels influence post-purchase satisfaction, we employed Model 4 of PROCESS 3.3 (Hayes) in SPSS 26.0, utilizing the Bootstrap method to assess the validity of the mediating effects of perceived information overload and flow experience.

In the mediated effects model, the engagement atmosphere was treated as a multi-category independent variable, while perceived information overload, flow experience, and post-purchase satisfaction were continuous variables. To test Hypotheses 6 and 7, the study employed both overall mediation and relative mediation analyze (Fang et al., 2017). The engagement atmosphere was coded into two dummy variables (X1 and X2). The overall total effect test yielded F (2,402) = 27.543, p < 0.001, indicating that at least one of the two relative total effects was non-zero. Specifically, the relative total effect for medium engagement atmosphere was −0.2172, p > 0.05, with a 95 % bootstrap confidence interval of [−0.5250,0.906], signifying insignificance. Conversely, the relative total effect for high engagement atmosphere was −1.2174, p < 0.001, with a 95 % bootstrap confidence interval of [−1.5719, −0.8629], confirming significance. The overall direct effect test (F (2,402) = 10.864, p < 0.001) further indicated that at least one relative direct effect was non-zero, necessitating further relative mediation analysis.

The relative mediation analyses yielded the following results. When using perceived information overload as the mediator and low engagement atmosphere as the reference level, in the medium engagement atmosphere, the relative mediation effect is significant (a1 = 0.3665, b = −0.1569, a1*b = −0.0575, 95 % CI: [−0.1331, −0.0056]). However, when flow experience was the mediator, the relative mediation effect is nonsignificant (95 % CI: [−0.0902, 0.0417]). The relative direct effect is nonsignificant (effect = −0.1377, p > 0.05, 95 % CI: [−0.4328, 0.1574]), and the relative total effect was also nonsignificant (c=−0.2172, p > 0.05).

For high engagement atmosphere, when using perceived information overload as the mediator, the relative mediation effect is significant (a1 = 1.2191, b = −0.1569, a1*b = −0.1912, 95 % CI: [−0.3374, −0.0689]). This indicates that a high engagement atmosphere increased perceived information overload by 1.2191, which led to decreased post-purchase satisfaction (b = −0.1569). Using flow experience as the mediator, the relative mediation effect is significant (a2 = 1.1399, b = −0.2079, a2*b = −0.2370, 95 % CI: [−0.3757, −0.1166]). A high engagement atmosphere enhanced the flow experience by 1.1399, which similarly reduced post-purchase satisfaction (b = −0.2079). The relative direct effect is significant (effect = −0.7892, p < 0.001, 95 % CI: [−1.1514, −0.4270]), indicating that even after accounting for mediation, high engagement atmosphere reduced post-purchase satisfaction by 0.7892 compared to low engagement atmosphere. The relative total effect was significant (c = −1.2174, p < 0.001), with mediation effects accounting for 35.2 % of this total ((0.1913 + 0.2370)/1.2174). Specific results are shown in Figs. 11 and 12 Table 7.

4.2.5. Test of moderating effects

To investigate whether the effect of engagement atmosphere on consumers' post-purchase satisfaction—mediated by perceived information overload and flow experience—is moderated by gender, we employed PROCESS (Model 14) in SPSS 26.0 (Fang and Wen, 2022).

The independent variable, engagement atmosphere, was a three-level categorical variable (low, medium, high) and dummy-coded with low engagement atmosphere as the reference level, generating two dummy variables (X1 and X2), while gender served as the moderator (male = 1, female = 2). For male participants, the relative mediation effect of medium engagement atmosphere via perceived information overload on post-purchase satisfaction was significant (Effect = −0.0649, SE = 0.0392, 95 % CI: [−0.1512, −0.0028]), but nonsignificant via flow experience (Effect = −0.0108, SE = 0.0187, 95 % CI:

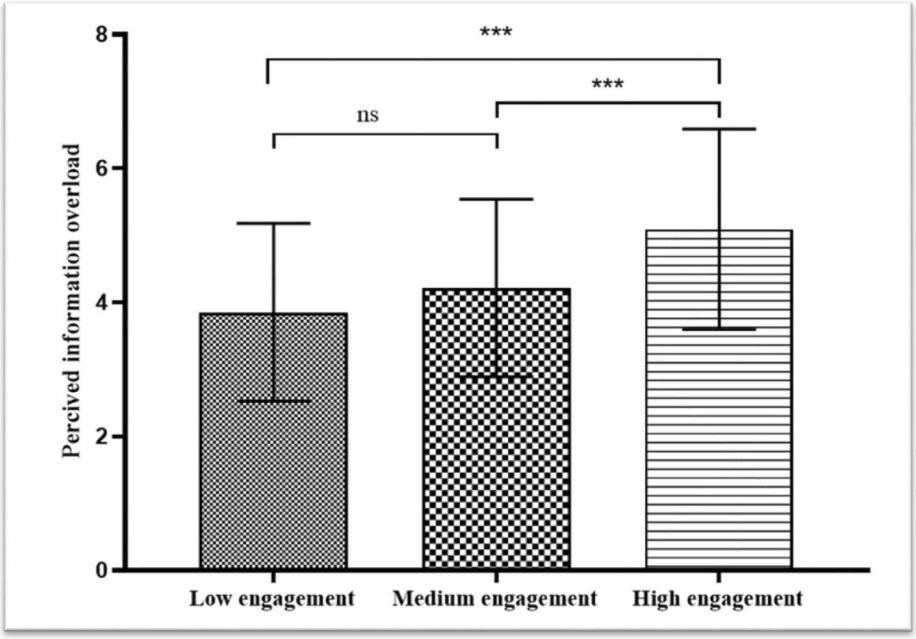


Fig. 8. Differences in the impact of engagement on perceived information overload.

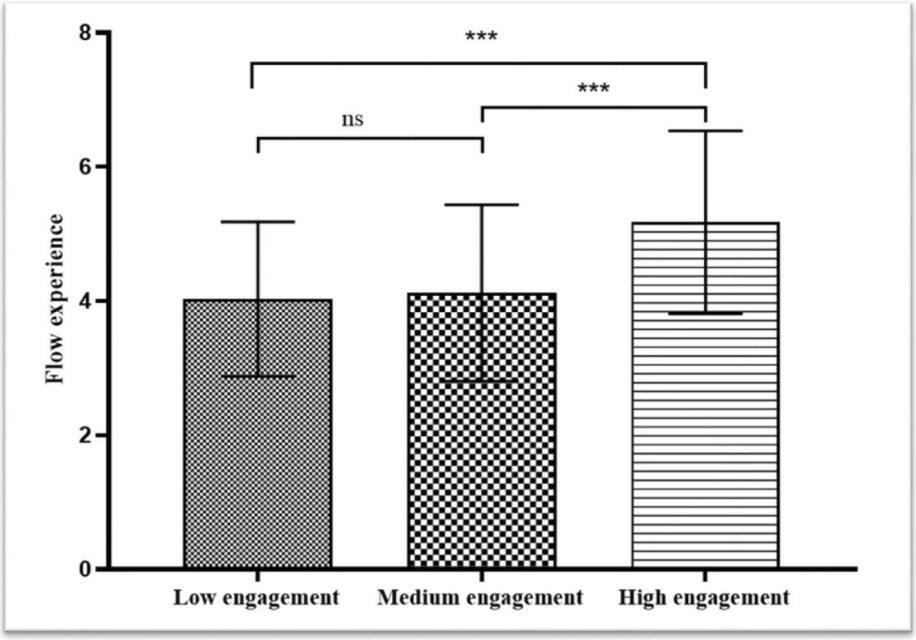


Fig. 9. Differences in the impact of engagement on flow experience.

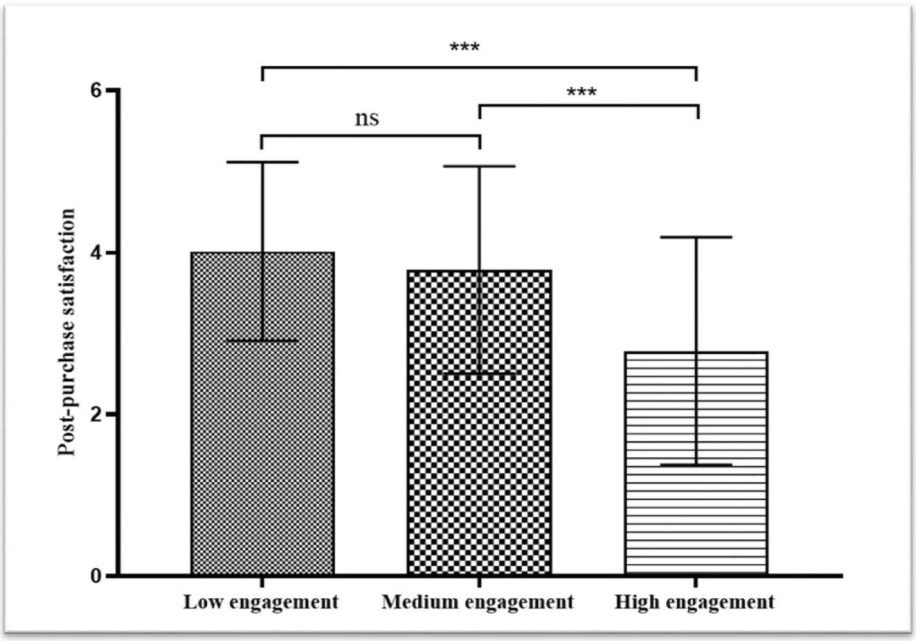


Fig. 10. Differences in the impact of engagement on post-purchase satisfaction.

[−0.0525,0.0254]). For female participants, the relative mediation effect of medium engagement atmosphere was nonsignificant via both perceived information overload (Effect = −0.0451, SE = 0.0328, 95 % CI: [−0.1215,0.0037]) and flow experience (Effect = −0.0368, SE = 0.0537, 95 % CI: [−0.1512,0.0682]). The difference in relative mediation effects between males and females for perceived information overload was nonsignificant (Effect = −0.0198, SE = 0.038, 95 % CI: [−0.1038,0.0529]), indicating that gender does not moderate the mediating effects of medium engagement atmosphere on post-purchase satisfaction relative to low engagement atmosphere.

For male participants, the relative mediation effect of high engagement atmosphere on post-purchase satisfaction through perceived information overload was significant (Effect = −0.2160, SE = 0.0907, 95 % CI: [−0.4006, −0.0450]), while the effect through flow experience was nonsignificant (Effect = −0.1160, SE = 0.0786, 95 % CI: [−0.2706,0.0377]). For female participants, the mediation effect via perceived information overload was nonsignificant (Effect = −0.1501, SE=0.0848, 95 % CI: [−0.3266,0.0031]), whereas the effect through flow experience was significant (Effect = −0.3955, SE = 0.1006, 95 % CI: [−0.6046, −0.2104]). Gender differences in mediation effects were nonsignificant for perceived information overload (Effect = −0.0660, SE = 0.1138, 95 % CI: [−0.2824,0.1662]) but significant for flow experience (Effect = 0.2795, SE = 0.1205, 95 % CI: [0.0665,0.5363), suggesting that gender moderates the mediation effect of high

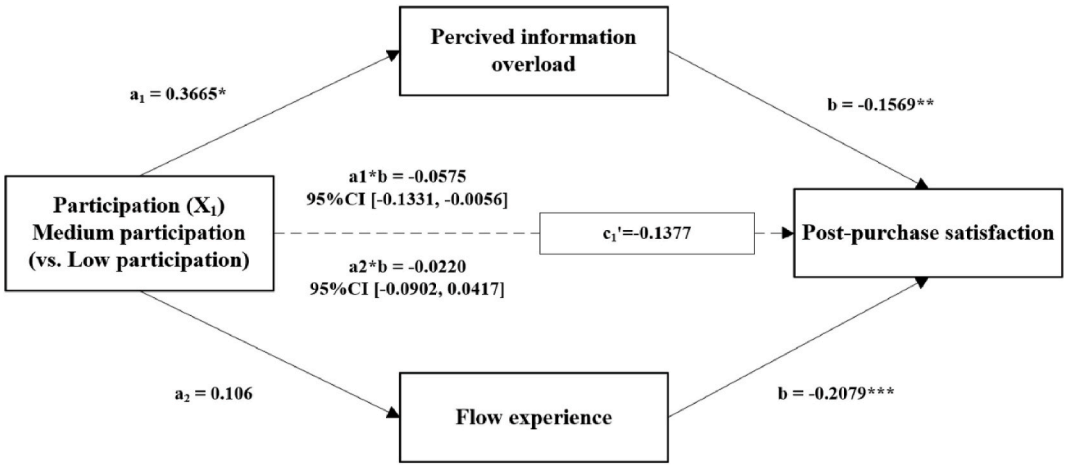


Fig. 11. Relative mediation effect results for medium engagement atmosphere (with low engagement atmosphere as a control).

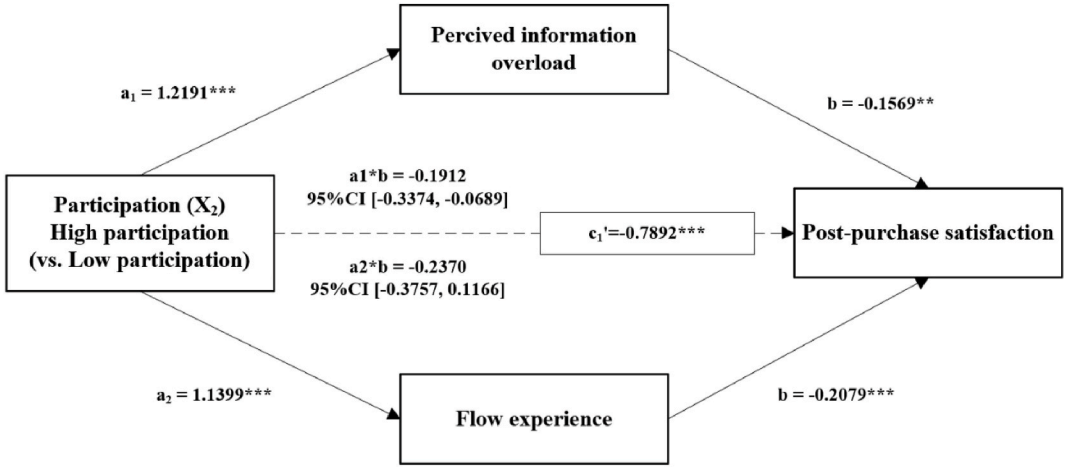


Fig. 12. Relative mediation effect results in a high engagement atmosphere (with a low engagement atmosphere as a control).

Table 7
Results of the mediating effect of perceived information overload and the flow experience (with reference to a low engagement atmosphere).

Mediation effect path	β	95 % CI		R^2
		LLCI	ULCI	
Medium-engagement → Perceived information overload → Post-purchase satisfaction	−0.0575	−0.1331	−0.0056	0.1060
High-engagement → Perceived information overload → Post-purchase satisfaction	−0.1912	−0.3374	−0.0689	
Medium-engagement → Flow experience → Post-purchase satisfaction	−0.022	−0.0902	0.0417	0.1243
High-engagement → Flow experience → Post-purchase satisfaction	−0.237	−0.3757	−0.1166	
Medium -engagement → Post-purchase satisfaction	−0.2172	−0.5250	0.0906	0.2092
High-engagement → Post-purchase satisfaction	−1.2174	−1.5719	−0.8629	

engagement atmosphere on satisfaction via flow experience. Specific results are shown in Table 8 and Fig. 13.

In summary, the findings supported Hypothesis 9 but not Hypothesis 8. Specifically, gender plays a moderating role between flow experience and post-purchase satisfaction in high-engagement contexts, with male reinforcing the negative relationship between the two. However, gender did not exhibit a moderating effect in either medium- or high-engagement contexts when perceived information overload served as the mediating variable.

4.2.6. *Robustness test*

To validate the robustness of the observed high engagement-low satisfaction paradox, engagement intensity was stratified using descending tertile segmentation. The reformed groups exhibited distinct profiles: low engagement (M=2.774, SD=0.520), moderate engagement (M=4.276, SD=0.462), and high engagement (M=6.030, SD=0.639). Following verification of normality and homoscedasticity, a one-way ANOVA revealed significant engagement effects on perceived information overload, flow experience, and post-purchase satisfaction ($p < 0.001$). High engagement contexts generated greater cognitive load ($M_{High} = 4.896$, $M_{Medium} = 4.224$, $M_{Low} = 3.918$, $p < 0.001$) and deeper flow states ($M_{High} = 4.960$, $M_{Medium} = 4.092$, $M_{Low} = 4.035$, $p < 0.001$), yet paradoxically reduced satisfaction ($M_{High} = 3.000$, $M_{Medium} = 3.847$, $M_{Low} = 3.922$, $p < 0.001$), thereby supporting H1- H3. Furthermore, the effects of perceived information overload and flow experience on consumer post-purchase satisfaction were found to be negatively significant (perceived information overload: $B = -0.1708$, $SE = 0.047$, $t = -3.6152$, $p < 0.001$, $LLCI = -0.2637$, $ULCI = -0.0779$; flow experience: $B = -0.2213$, $SE = 0.050$, $t = -4.4093$, $p < 0.001$, $LLCI = -0.3199$, $ULCI = -0.1226$), thereby supporting hypotheses H4 and H5.

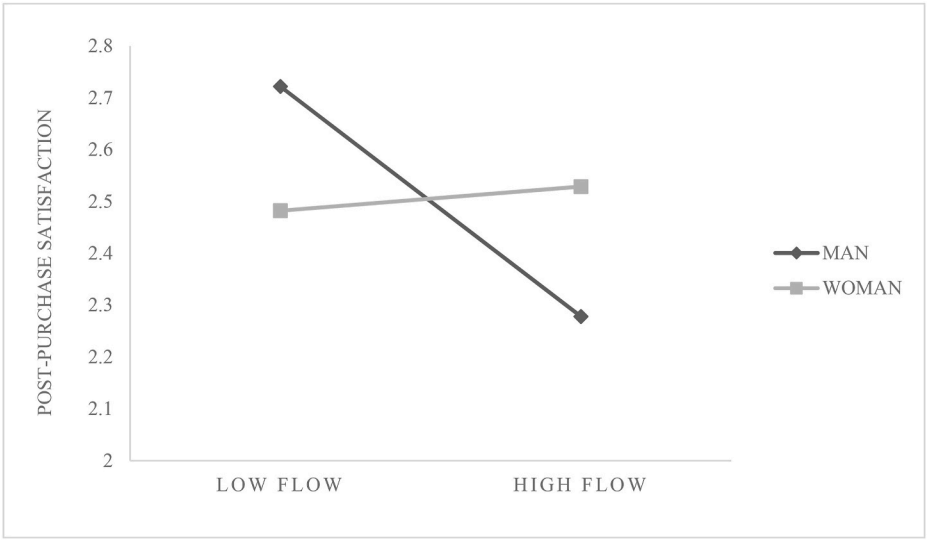


Fig. 13. Simple slope plot of the moderating effect.

Table 8
Results of the analysis of the mediating effect of the three categories of independent variables being moderated.

Predictive variables	Perceived information overload (M1)		Flow experience (M2)		Post-purchase satisfaction (Y)	
	β	t	β	t	β	t
X1 (Medium vs Low)	0.3665	2.1772*	0.1060	0.6699	−0.1363	−0.9115
X2 (High vs Low)	1.2191	6.2884***	1.1399	6.2554***	−0.7844	−4.2752***
Gen					0.0055	0.0454
M1					−0.1507	−3.2047**
M2					−0.2219	−4.4049***
M1*GEN					−0.0541	−0.5852
M2*GEN					0.2452	2.515*
R^2	0.106		0.1243		0.2222	
F	11.7674		14.0892		12.4403	

Note:*p < 0.05, **p < 0.01, ***p < 0.001.

The mediation analysis revealed differential effects across engagement atmosphere levels. When using the low-engagement atmosphere as the reference: Under medium-engagement conditions, neither perceived information overload (95 % CI [-0.1267, 0.0012]) nor flow experience (95 % CI [-0.0880, 0.0552]) demonstrated significant relative mediation effects. Both direct and total effects remained non-significant. In high-engagement atmospheres, significant relative mediation effects emerged through both perceived information overload (95 % CI [-0.2845, −0.0656]) and flow experience (95 % CI [-0.3246, −0.1000]). The direct effect (95 % CI [-0.8530, −0.2259]) and total effect (95 % CI [-1.2209, −0.5944]) both reached statistical significance, with a relative mediation effect size of 40.6 %. Complete numerical results are presented in Figs. 14 and 15.

The moderation analysis of gender effects revealed distinct patterns across engagement conditions: Under medium - engagement atmospheres, no significant gender moderation was observed on the relative

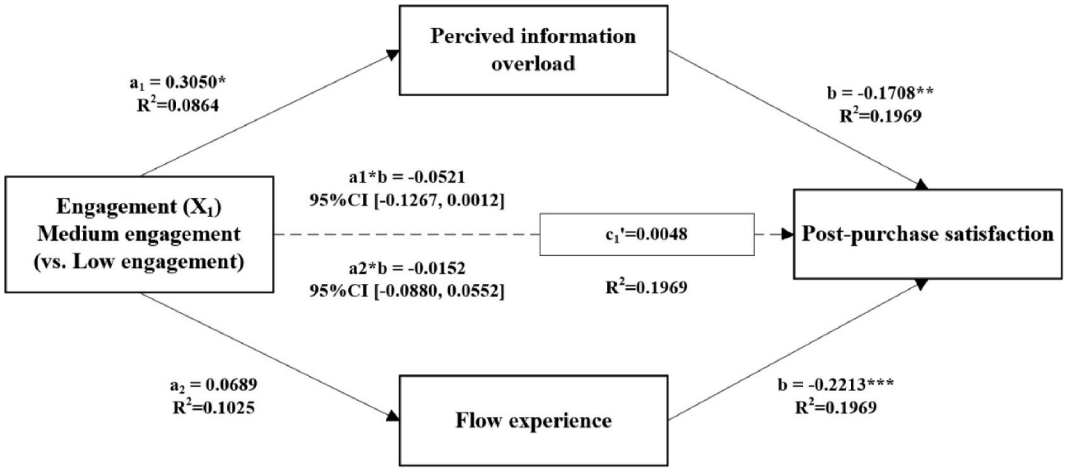


Fig. 14. Relative mediation effect results for medium engagement atmosphere (with low engagement atmosphere as a control).

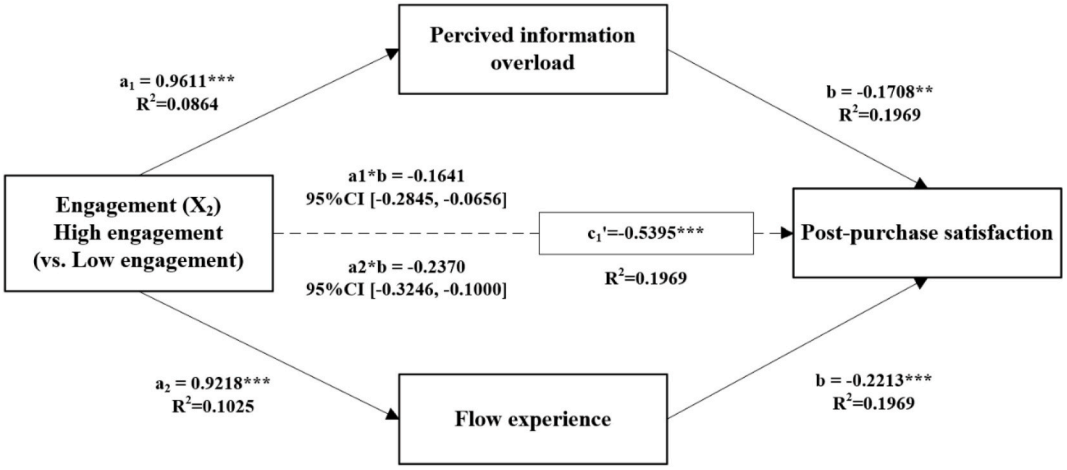


Fig. 15. Relative mediation effect results for high engagement atmosphere (with low engagement atmosphere as a control).

mediation effects (vs. low engagement), regardless of the mediator; In high-engagement contexts, gender moderation emerged only when flow experience served as the mediator. Specifically, for males, the relative mediation effect through flow experience was significant (β = −0.3366, SE = 0.0878, 95 % CI [-0.5199, −0.1756]). For females, this mediation pathway remained non-significant (β = −0.1017, SE = 0.0642, 95 % CI [-0.2378, 0.0141]). This indicates that male consumers experiencing flow exhibited stronger reductions in post-purchase satisfaction compared to females, suggesting gender-specific vulnerability to immersive engagement outcomes. Specific results are shown in Table 9 and Fig. 16.

The test results showed that the negative relationship, mediating effect, and moderating effect were significantly established, indicating that the inhibition and influence mechanisms found in the previous stage were not coincidental and had strong stability and reliability.

4.2.7. Supplementary analyses

To explore gender differences in depth, the study considered the interaction of gender with other covariates (frequency/duration) while controlling for age and occupation and followed the methodological process in section 4.2.5 for testing.

4.2.7.1. Moderation effects of gender-viewing frequency interaction. Under medium engagement, no significant moderation occurred in the relative mediation pathway via information overload and flow experience. In high-engagement contexts, significant gender-frequency interaction (β = −0.3735, SE = 0.0970, 95 % CI: [-0.5735, −0.1948]) for low-frequency males, versus non-significance in high-frequency females. The gender-stratified effect disparity reached significance (Δβ = 0.2551, SE = 0.1205, 95 % CI: [0.0415, 0.5105]), indicating males with infrequent viewing exhibit amplified flow-induced dissatisfaction in high-engagement environments. Specific results are shown in Table 10 and Fig. 17.

4.2.7.2. Moderation effects of gender-viewing duration interaction. Under medium engagement, no significant moderation occurred in the relative mediation pathway via information overload and flow experience. In high-engagement contexts, significant gender - viewing duration interaction (β = −0.3735, SE = 0.0970, 95 % CI: [-0.5735, −0.1948]) for short -duration males, versus non-significance in long-duration females. The gender-stratified effect disparity reached significance (Δβ = 0.2551, SE = 0.1205, 95 % CI: [0.0415, 0.5105]). This duration-dependent gender asymmetry echoes gender schema theory and cognitive load theory, suggesting time-constrained males experience accelerated resource depletion during high-engagement consumption. Specific results are shown in Table 11 and Fig. 18.

4.3. Study3

4.3.1. Data collection

This study replicated Study 2's measurement instruments and procedures for assessing engagement, perceived information overload, flow experience, and post-purchase satisfaction. To enhance cross-cultural validity and methodological rigor, we incorporated a multinational comparative design, recruiting additional North American and Asia-Pacific participants via Credamo, a mainstream crowdsourcing platform. After eliminating duplicates, outliers, and speeders, 320 valid responses (86.49 % validity rate) were retained with complete data. Table 12 provides detailed statistics.

Table 9
Results of the analysis of the mediating effect of the three categories of independent variables being moderated.

Predictive variables	Perceived information overload (M1)		Flow experience (M2)		Post-purchase satisfaction (Y)	
	β	t	β	t	β	t
X1 (Medium vs Low)	0.3050	1.7908	0.0689	0.4297	0.0079	0.0526
X2 (High vs Low)	0.9611	5.6398***	0.9218	5.7463***	−0.5446	−3.4319**
GEN					0.0200	0.1642
M1					−0.164	−3.4809**
M2					−0.2352	−4.6712***
M1*GEN					−0.0461	−0.4954
M2*GEN					0.2549	2.5960***
R^2	0.0864		0.1025		0.2112	
F	9.3892		11.3334		11.6622	

Note:*p < 0.05, **p < 0.01, ***p < 0.001.



Fig. 16. Simple slope plot of the moderating effect.

Table 10
Results of the analysis of the mediating effect of the three categories of independent variables being moderated.

Predictive variables	Perceived information overload (M1)		Flow experience (M2)		Post-purchase satisfaction (Y)	
	β	t	β	T	β	t
X1 (Medium vs Low)	0.3665	2.1772	0.1060	0.6699	−0.1334	−0.8900
X2 (High vs Low)	1.2191	6.2884*	1.1399	6.2554***	−0.7758	−4.2185***
GEN-FRE					0.0114	0.3302
M1					−0.1542	−3.2767**
M2					−0.2190	−4.3432***
M1*GEN-FRE					−0.0230	−0.8832
M2*GEN-FRE					0.0648	2.2727*
R^2	0.1060		0.1243		0.2199	
F	11.7674		14.0892		12.2802	

Note:*p < 0.05, **p < 0.01, ***p < 0.001.

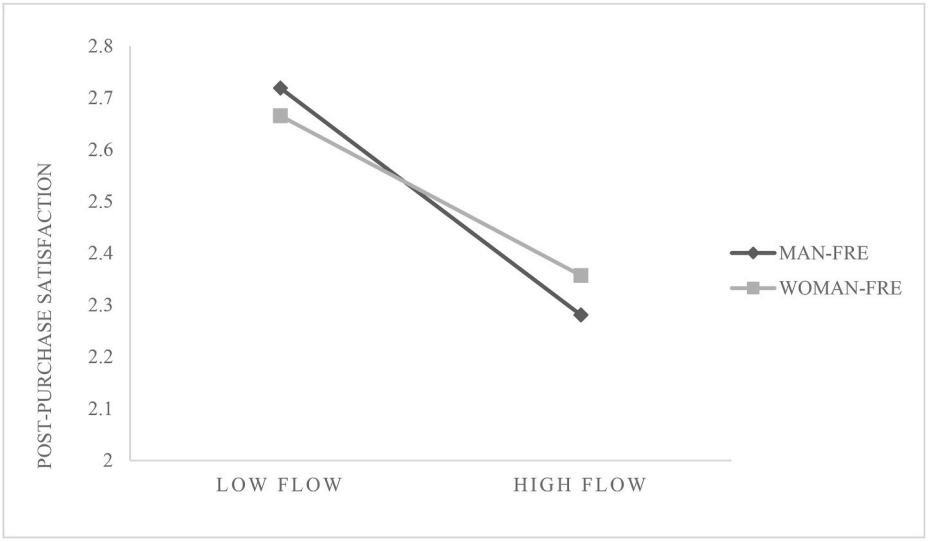


Fig. 17. Simple slope plot of the moderating effect.

4.3.2. Reliability and validity analysis

4.3.2.1. Reliability test. The reliability of the sample data was assessed using SPSS 26.0, with the results detailed in Table 13. Cronbach's α values for each variable ranged from 0.828 to 0.865, this indicates that the data from the scale is reliable. Collinearity diagnostics revealed acceptable VIF ranges (1.185–1.280), significantly below the 5.0 threshold. Durbin-Watson analysis confirmed residual independence (DW=1.990).

Table 11
Results of the analysis of the mediating effect of the three categories of independent variables being moderated.

Predictive variables	Perceived information overload (M1)		Flow experience (M2)		Post-purchase satisfaction (Y)	
	β	t	β	t	β	t
X1 (Medium vs Low)	0.3665	2.1772	0.1060	0.6699	−0.1383	−0.9252
X2 (High vs Low)	1.2191	6.2884***	1.1399	6.2554***	−0.7898	−4.3037***
GEN-TIME					−0.0491	−1.0853
M1					−0.1544	−3.2881**
M2					−0.2132	−4.2437***
M1*GEN-TIME					0.0032	0.0913
M2*GEN-TIME					0.0804	2.1808*
R^2	0.1060		0.1243		0.2233	
F	11.7674		14.0892		12.5189	

Note:*p < 0.05, **p < 0.01, ***p < 0.001.

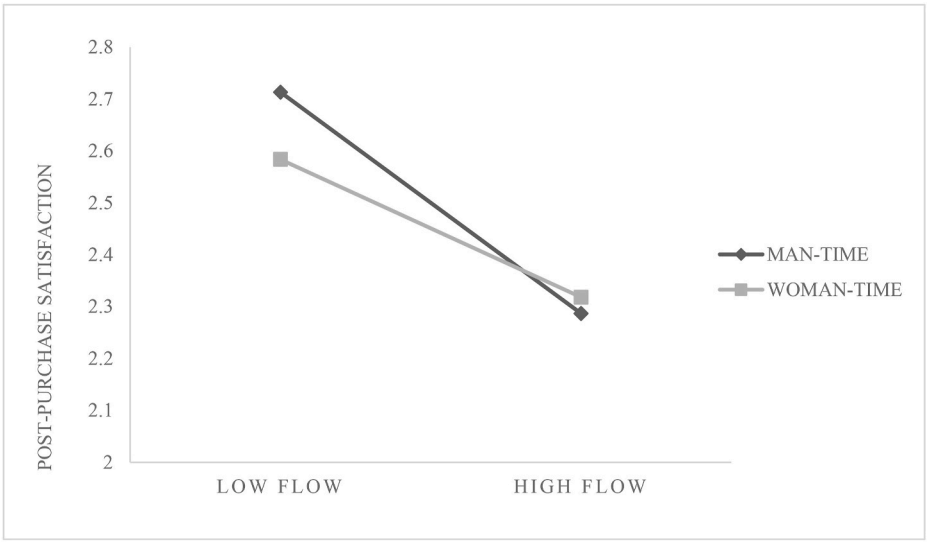


Fig. 18. Simple slope plot of the moderating effect.

4.3.2.2. Validity test. Average variance extracted (AVE) values for all variables ranged from 0.677 to 0.699, and composite reliability (CR) values ranged from 0.870 to 0.894. These results indicate that the measurement indicators had strong explanatory power for each variable. Furthermore, Table 14 presents the discriminant validity indices, showing significant correlations among the four variables ($p < 0.01$). The coefficient values were lower than the square roots of the AVE, indicating satisfactory discriminant validity between the variables.

4.3.2.3. Common method bias. To assess common method bias, Harman's single-factor test was applied. Results showed that the largest principal component accounted for 28.465 % of the total variance, well below the 40 % threshold. This confirms the absence of significant methodological factors influencing the variance in this study. In addition, we tested for common method bias using single-factor confirmatory factor analysis (CFA). The result of the CFA test shows that x2/df = 1.201, RMSEA = 0.025, NFI = 0.967, IFI = 0.994, TLI = 0.990, CFI = 0.994, while the single-factor CFA's fit index results are x2/df = 17.364, RMSEA = 0.225, NFI = 0.584, IFI = 0.598, TLI = 0.360, CFI = 0.593, the fit index of one-factor CFA is much poorer than the original model, which indicates that the validity of this model passes the test.

4.3.3. Main effect test

Following the same methodological process as in 4.2.3, we categorized engagement into high (M = 6.292, SD = 0.504), medium (M = 4.288, SD = 0.657), and low levels (M = 2.563, SD = 0.455). Normality and homogeneity of variance assumptions were verified (Appendix B). One-way ANOVA revealed significant engagement effects on perceived information overload (PIO), flow experience, and post-purchase satisfaction (all $p < 0.001$). High engagement generated elevated PIO ($M_{High} = 5.025$, $M_{Medium} = 4.171$, $M_{Low} = 3.787$, $p < 0.001$) and FL ($M_{High} = 5.028$, $M_{Medium} = 4.141$, $M_{Low} = 4.098$, $p < 0.001$), but paradoxically reduced PS ($M_{High} = 2.806$, $M_{Medium} = 3.866$, $M_{Low} = 4.025$, $p < 0.001$), reconfirming H1-H3. Path analyses demonstrated significant negative impacts on satisfaction: PIO (B = −0.1845, SE = 0.0531, t (320) = −3.47, $p < 0.01$, 95 % CI [-0.289, −0.080]); flow experience (B = −0.1936, SE = 0.0561, t (320) = −3.45, $p < 0.01$, 95 % CI [-0.304, −0.083]), thereby validating H4-H5. The counterintuitive satisfaction suppression under high engagement aligns with cognitive-affective conflict theory, where heightened immersion amplifies post-consumption dissonance. Specific results are shown in Figs. 19, 20 and 21.

4.3.4. Mediated effects test

Following the relative mediation effect analysis in 4.2.4, we got the relative mediation effect results here. Taking the low-engagement atmosphere as a reference, in the medium-engagement atmosphere, the relative mediation effect was significant with perceived information overload as the mediator but not significant with the heart flow experience as the mediator. The relative direct and total effects were also not significant (c = −0.1565, $p > 0.05$). For the high-engagement atmosphere, using the low-engagement atmosphere as a reference, the relative mediation effects were significant whether the mediator was perceived as information overload or the flow experience. The relative direct effect was significant (Effect = −0.7817, $p < 0.01$, 95 % CI: [-1.1882, −0.3752]), the relative total effect was significant (C =

Table 12
Demographic characteristics of sample.

Classification	Features	Number	Proportion
Gender	Man	163	50.94 %
	Woman	157	49.06 %
Age	Under 18 years old	15	4.69 %
	18–25	87	27.19 %
	26–30	82	25.63 %
	31–40	88	27.50 %
	41–50	30	9.38 %
	51–60	15	4.69 %
	Above 60 years old	3	0.94 %
Occupation	Students	99	30.94 %
	State-owned Enterprises	56	17.50 %
	Institutions	55	17.19 %
	Civil Servants	57	17.81 %
	Private enterprises	41	12.81 %
	Foreign Invested Enterprises	12	3.75 %
Monthly income	< 822 USD	94	29.38 %
	823-1370 USD	80	25.00 %
	1371-2056 USD	77	24.06 %
	> 2057 USD	69	21.56 %
Frequency of viewing live sales	Never	0	0.00 %
	1-2 times per month	105	32.81 %
	1-2 times per week	106	33.13 %
	3-5 times per week	80	25.00 %
	Daily	29	9.06 %
The average time spent watching live streaming	Less than 30 min	102	31.88 %
	30 min - 1 h	108	33.75 %
	1 h - 2 h	85	26.56 %
	More than 2 h	25	7.81 %
Area	North America	135	42.19 %
	Asia Pacific (including East Asia, Southeast Asia and Oceania)	185	57.81 %

Table 13
Reliability and validity analysis of variables.

Variables	Measurements	Factor loading	AVE	CR	Cronbach's α
Perceived information overload (PIO)	PIO 1	0.815	0.699	0.874	0.829
	PIO 2	0.840			
	PIO 3	0.846			
Flow experience (FL)	FL 1	0.791	0.690	0.870	0.828
	FL 2	0.809			
	FL 3	0.889			
Post-purchase satisfaction (PS)	PS 1	0.790	0.677	0.894	0.865
	PS 2	0.848			
	PS 3	0.837			
	PS 4	0.816			

Table 14
Discriminant validity test of variables.

	1	2	3
Perceived information overload (PIO)	***0.836***		
Flow experience (FL)	0.416**	0.831	
Post-purchase satisfaction (PS)	−0.345**	−0.353**	0.829

Note: ** means that at the 0.01 level (two-tailed), the correlation is significant.

−1.2190, $p < 0.001$), and the relative mediation effect size was 35.9 %. Figs. 22 and 23 show the detailed data.

4.3.5. Test of moderating effects

Following the moderated effects testing in 4.2.5, we analyzed the results. We found that gender did not moderate the relative mediation effect of medium engagement compared to the low - engagement atmosphere. However, it did moderate the relative mediation effect of high engagement relative to the low - engagement atmosphere. For male consumers, with the low - engagement atmosphere as a reference, the relative mediation effect of high engagement was insignificant when perceived information overload was the mediator (Effect = −0.1570, SE = 0.1008, 95 % CI: [-0.3717, 0.0211]), but significant when the heart - flow experience was the mediator (Effect = −0.3663, SE = 0.1083, 95 % CI: [-0.5954, −0.1713]). For female consumers, the high - engagement relative mediation effect was significant when perceived information overload was the mediator (Effect = −0.2825, SE = 0.1061, 95 % CI [-0.5081, −0.0866]) and insignificant when the flow experience was the mediator (Effect = −0.0819, SE = 0.0796, 95 % CI [-0.2439, 0.0755]). The difference in mediating effects between genders was not significant for perceived information overload ($\Delta\beta$ = -0.1255, SE=0.1333, 95 % CI: [-0.3971,0.1431]) but significant for the flow experience ($\Delta\beta$ = -0.2844, SE=0.1290, 95 % CI: [0.0586,0.5639]). Thus, Hypothesis 9 was supported, while Hypothesis 8 remained untested. Specific results are shown in Table 15 and Fig. 24.

5. Conclusion

5.1. Discussion

This study combines live-streaming data analysis and questionnaire surveys to investigate the impact and mechanisms of different engagement atmospheres in live-streaming on consumers' post-purchase satisfaction. Convergent results from mixed methods indicate that while high-engagement atmospheres are often associated with strong purchase intentions, they paradoxically reduce post-purchase satisfaction (Liu et al., 2018; Chung et al., 2023; Zheng et al., 2023; Feng et al., 2024; Zhang and Zhang, 2024). Studies 2 and 3 systematically explore the underlying mechanisms. Grounded in expectation confirmation theory, information overload theory, and flow theory, we reveal dual mediation effects through information overload and flow experience. Convergent findings from Studies 2–3 demonstrate that in high-engagement live-streaming environments: Herd effects induced by intense social interactions elevate consumers' pre-purchase expectations (Feng et al., 2024), leading to significant expectation-confirmation gaps after product receipt. Information overload from excessive interaction intensity increases cognitive load, resulting in decision-making difficulties and reduced decision quality (Zhang and Ruan, 2024). Flow experiences immerse consumers in emotional states, overriding objective product evaluations and promoting affect-driven purchases. Post-experience emotional withdrawal further exacerbates dissatisfaction (Rook, 1987; Huang, 2016).

Gender schema theory explains moderating effects: Males exhibit stronger flow-mediated dissatisfaction under high engagement due to cognitive-emotional processing differences. While males typically employ rational decision-making, flow states in high-engagement contexts override their cognitive preferences, creating decision-behavior conflicts that trigger regret (Dittmar et al., 2004; Meyers-Levy, 1988, 1991).

5.2. Theoretical implications

This study makes three principal theoretical contributions to existing literature. First, grounded in expectation confirmation theory, we extend the scope of engagement research. Prior literature predominantly emphasizes the positive impacts of engagement on in-stream consumer behaviors, such as purchase intention (Vivek et al., 2012; Alhabash et al., 2015), trust-building with streamers (Liu et al., 2018), and loyalty enhancement (Nadeem et al., 2021). Our work broadens engagement to a holistic "engagement atmosphere" dimension, specifically identifying high-engagement contexts. We pioneer in revealing the dark-side effects of high-engagement atmospheres on post-stream purchase satisfaction.

Second, building on the ELM framework, we propose a dual-process mechanism demonstrating the parallel mediating roles of information

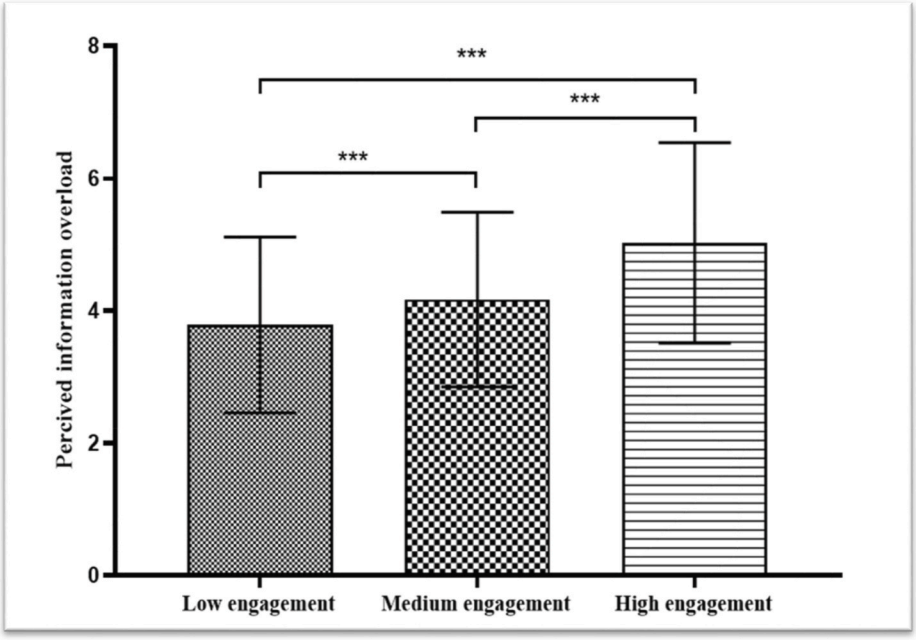


Fig. 19. Differences in the impact of engagement on perceived information overload.

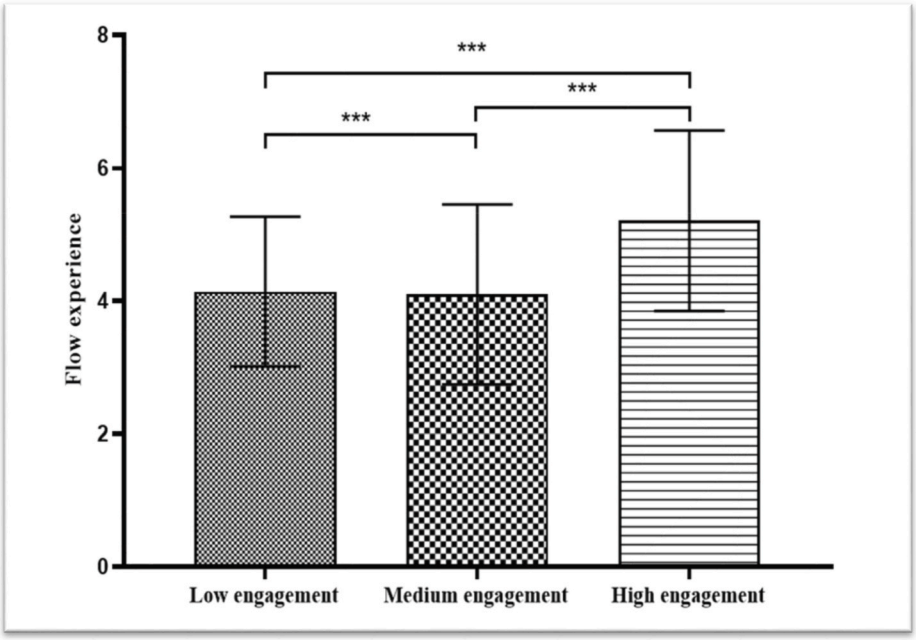


Fig. 20. Differences in the impact of engagement on flow experience.

overload and flow experience in high-engagement environments. By analyzing viewer-streamer and viewer-viewer interactions (Chen et al., 2022b), we expand our understanding of live-stream engagement, emphasizing its dual impact through informational (Zhao et al., 2023) and affective pathways. Particularly under high-interactivity conditions, information overload, and flow states coexist, creating a consumption paradox where conflicting affective responses alternately influence satisfaction. This integrates opposing perspectives: while prior studies predominantly associate information overload with reduced satisfaction (Liang et al., 2006),and flow with enhanced satisfaction (Sarwar et al., 2024), our synthesized framework advances a comprehensive understanding of multifaceted behavioral outcomes.

Third, applying gender schema theory, we uncover gender's moderating role in flow-satisfaction relationships, revealing divergent emotional reactions between genders in live-stream shopping. This extends gender schema theory to emerging digital marketing contexts and offers novel perspectives for future research. Practically, these findings provide a theoretical grounding for leveraging gender differences in live-stream marketing strategies, particularly in designing gender-adaptive engagement tactics (Meyers-Levy, 1988; Dittmar et al., 2004).

5.3. *Practical implications*

This study yields critical managerial implications for live-streaming

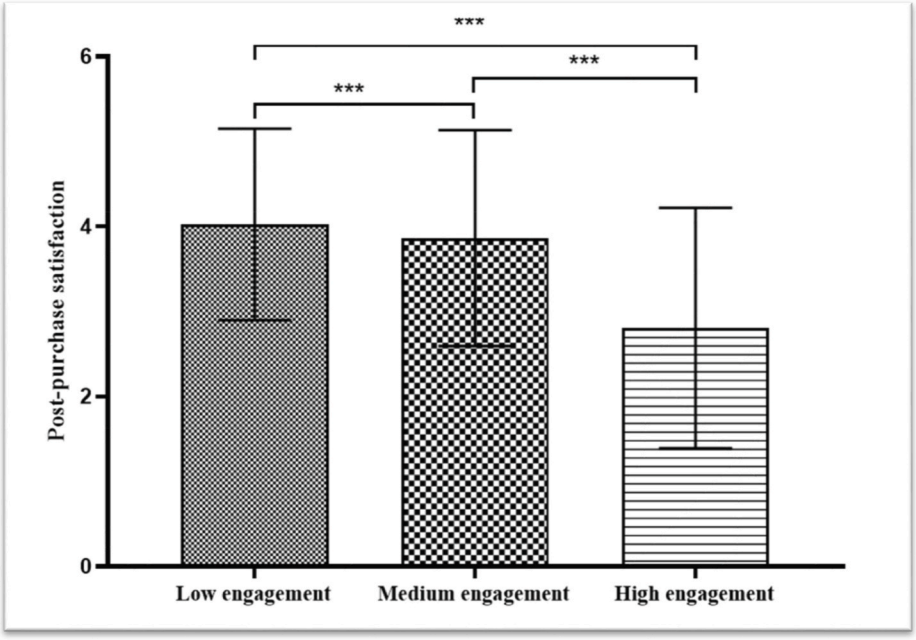


Fig. 21. Differences in the impact of engagement on post-purchase satisfaction.

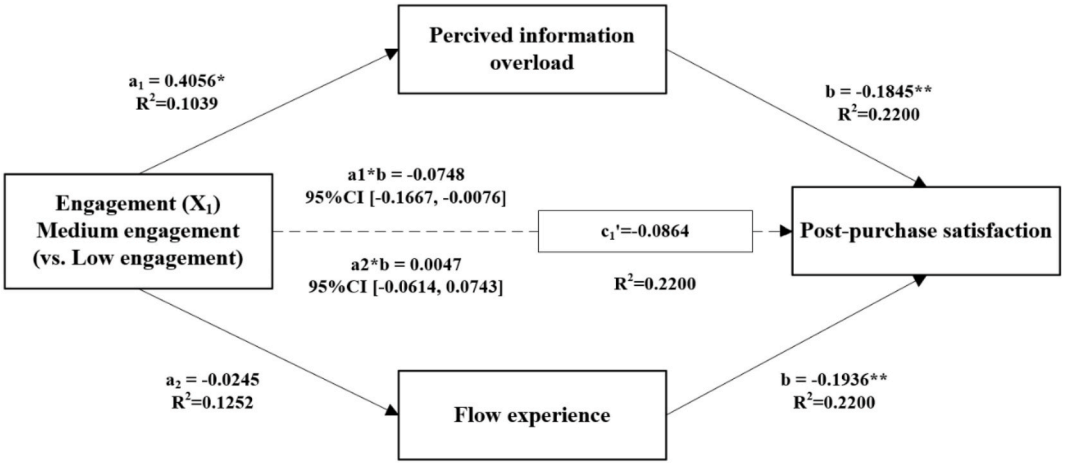


Fig. 22. Relative mediation effect results for medium engagement atmosphere (with low engagement atmosphere as a control).

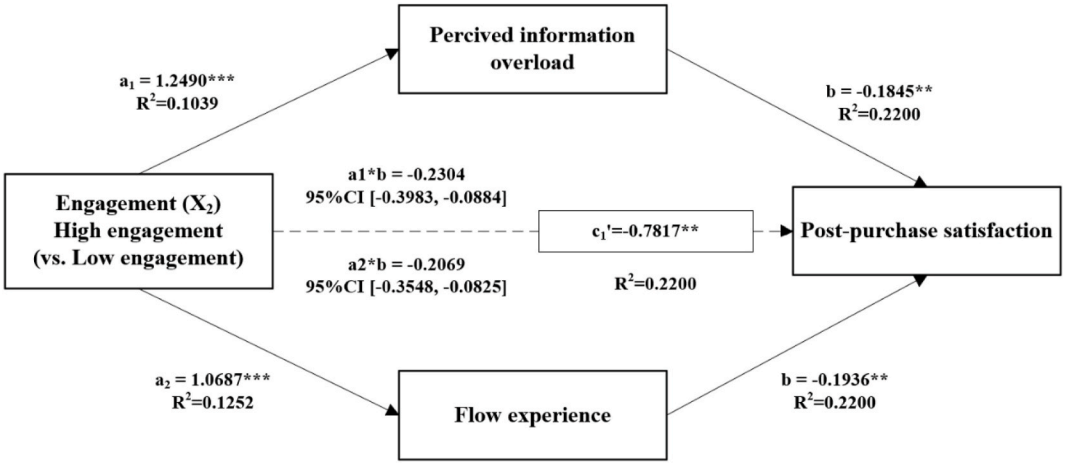


Fig. 23. Relative mediation effect results in a high engagement atmosphere (with a low engagement atmosphere as a control).

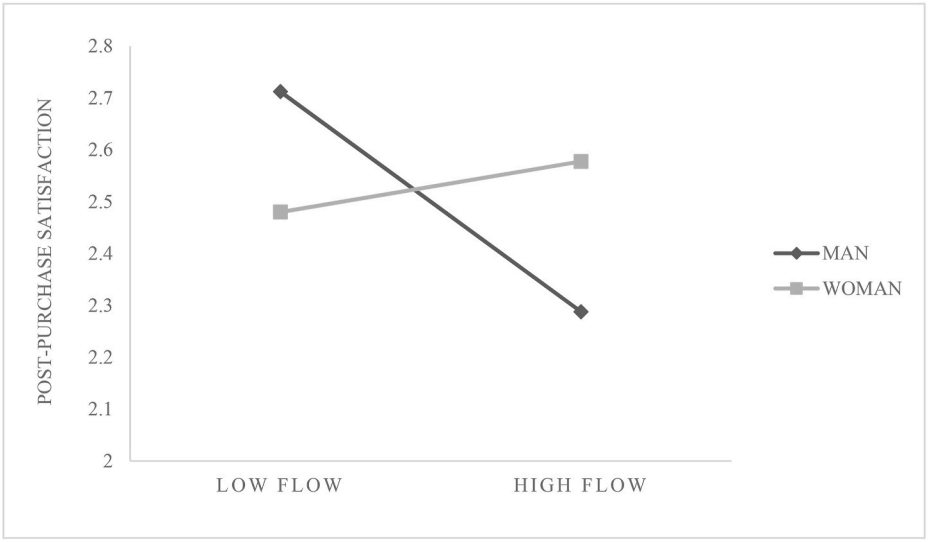


Fig. 24. Simple slope plot of the moderating effect.

Table 15
Results of the analysis of the mediating effect of the three categories of independent variables being moderated.

Predictive variables	Perceived information overload (M1)		Flow experience (M2)		Post-purchase satisfaction (Y)	
	β	t	β	t	β	t
X1 (Medium vs Low)	0.4056	2.1425*	−0.0245	−0.1365	−0.0664	−0.3939
X2 (High vs Low)	1.2490	5.7013***	1.0687	5.1503***	−0.7538	−3.661***
GEN					0.0289	0.2126
M1					−0.1750	−3.2974**
M2					−0.2122	−3.7559***
M1*GEN					−0.1005	−0.9669
M2*GEN					0.2661	2.4483*
R^2	0.1039		0.1252		0.2350	
F	9.1321		11.2744		10.5784	

Note:*p < 0.05, **p < 0.01, ***p < 0.001.

commerce. First, while high-engagement atmospheres effectively attract consumers, their adverse effects require strategic mitigation. Brand managers and streamers should avoid pursuing excessive participation intensity. Instead, they should balance product demonstrations with moderated engagement enhancements, particularly reducing redundant interactions and information repetition (Zhang and Ruan, 2024). Minimizing abrupt surges in interaction frequency and informational density could alleviate perceived information overload, thereby improving post-purchase satisfaction. AI-driven real-time analytics could optimize this balance by monitoring consumer cognitive loads.

Second, although flow experience strongly correlates with impulse purchases, over-immersion in high-engagement contexts may trigger suboptimal purchase decisions that compromise post-purchase satisfaction. Practitioners should design content to counteract emotional dominance by integrating educational components (e.g., product value explanations usage scenarios) and avoiding hyperbolic claims. This balanced approach reduces post-purchase dissonance and return rates while maintaining engagement efficacy.

Third, gender-based content customization enhances strategy precision. For male consumers, streamlined interactions in high-engagement environments minimize post-flow negative spillovers by prioritizing goal-oriented purchasing processes. Conversely, female audiences benefit from sustained interactive formats that preserve flow experience benefits. Such differentiated implementations of engagement mechanics can optimize satisfaction outcomes across demographics.

Finally, regulatory frameworks must evolve to address platform governance. Key measures include: 1) Establishing industry standards for information transparency (e.g., clear labeling of promotional content), 2) Limiting concurrent promotional campaigns to prevent participation fatigue, and 3) Implementing accountability mechanisms to penalize deceptive practices. These policies would foster sustainable ecosystem development while protecting consumer rights.

Collectively, these recommendations advocate for a nuanced engagement strategy that harmonizes cognitive load management, flow experience regulation, and demographic responsiveness—a triad essential for optimizing long-term consumer satisfaction in live-streaming commerce.

5.4. Limitations and prospects

While this study offers valuable insights, several limitations warrant future investigation. First, although we identified mediating and moderating mechanisms between engagement atmospheres and post-purchase satisfaction, the complexity of underlying processes requires deeper exploration. Contextual factors like product types and streamer influence may differentially shape these relationships.

Methodologically, two constraints emerge: 1) While Study 1 utilized actual live streaming data, its limited scope necessitates refined operationalization of engagement metrics (e.g., quantifying participation behaviors) and satisfaction measurement (e.g., NLP-based sentiment analysis of reviews). 2) Study 2's retrospective survey design could be supplemented with longitudinal experiments tracking real-time decision phases.

Additionally, sample constraints prevent conclusive examination of potential curvilinear effects between engagement intensity and satisfaction across scenarios. Future research should employ diversified

methods (e.g., multi-wave field experiments, cross-cultural sampling) to disentangle these nonlinear dynamics.

CRediT authorship contribution statement

Xiaojun Fan: Writing – review & editing, Conceptualization. **Wenxia Tao:** Writing – original draft, Data curation. **Haowei Yu:** Resources. **Zheng Wang:** Resources.

Declaration of competing interest

All authors disclosed no relevant relationships.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jretconser.2025.104303.

Data availability

Data will be made available on request.

References

Alhabash, Saleem, McAlister Anna, R., Lou, Chen, Amy, Hagerstrom, 2015. From clicks to behaviors: the mediating effect of intentions to like, share, and comment on the relationship between message evaluations and offline behavioral intentions. J. Interact. Advert. 15 (2), 82–96.
Ameen, N., Tarhini, A., Shah, M.H., Madichie, N.O., 2020. Employees' behavioural intention to smartphone security: a gender-based, cross-national study. Comput. Hum. Behav. 104, 14.
Barger, V., Peltier, J.W., Schultz, D.E., 2016. Social media and consumer engagement: a review and research agenda. J. Res. Indian Med. 10 (4), 268–287.
Busalim, A.H., Ghabban, F., Hussin, A.C., 2021. Customer engagement behaviour on social commerce platforms: an empirical study. Technol. Soc. 64, 17.
Cacioppo, John T., Petty, Richard E., Feng, Kao Chuan, Regina, Rodriguez, 1986. Central and peripheral routes to persuasion: an individual difference perspective. J. Pers. Soc. Psychol. 51 (5), 1032.
Calvo-Porral, Cristina, Agustín, Ruiz-Vega, Jean-Pierre, Lévy-Mangin, 2021. How consumer involvement influences consumption-elicited emotions and satisfaction. Int. J. Mark. Res. 63 (2), 251–267.
Chen, Jialiang, Yin, Wu, 2024. Would you be willing to purchase virtual gifts during esports live streams? Streamer characteristics and cultural traits. Comput. Hum. Behav. 152, 108075.
Chen, Y.C., Shang, R.A., Kao, C.Y., 2009. The effects of information overload on consumers' subjective state towards buying decision in the internet shopping environment. Electron. Commer. Res. Appl. 8 (1), 48–58.
Chen, C.D., Zhao, Q., Wang, J.L., 2022a. How livestreaming increases product sales: role of trust transfer and elaboration likelihood model. Behav. Inf. Technol. 41 (3), 558–573.
Chen, H.Q., Zhang, S.H., Shao, B.J., Gao, W., Xu, Y.J., 2022b. How do interpersonal interaction factors affect buyers' purchase intention in live stream shopping? The mediating effects of swift guanxi. Internet Res. 32 (1), 335–361.
Chia-Chen, Chen, Kuo-Lun, Hsiao, Shan-Jung, Wu, 2018. Purchase intention in social commerce. Libr. Hi Tech 36 (4), 583–604.
Chung, J.Y., Ding, Y., Kalra, A.I., 2023. Really know you: how influencers can increase audience engagement by referencing their close social ties. J. Consum. Res. 50 (4), 683–703.
Churchill, Gilbert A., Carol, Surprenant, 1982. An investigation into the determinants of customer satisfaction. J. Mark. Res. 19 (4), 491–504.
Clement, Addo Prince, Fang, Jiaming, Andy, Ohemeng Asare, Bakabbey, Kulbo Nora, 2021. Customer engagement and purchase intention in live-streaming digital marketing platforms. Serv. Ind. J. 41 (11–12), 767–786.
Csikszentmihalyi, Mihaly, 2000. Beyond Boredom and Anxiety. Jossey-bass.
Dittmar, Helga, Karen, Long, Rosie, Meek, 2004. Buying on the internet: gender differences in on-line and conventional buying motivations. Sex. Roles 50 (5), 423–444.
Diwanji, V., Reed, A., Ferchaud, A., Seibert, J., Weinbrecht, V., Sellers, N., 2020. Don't just watch, join in: exploring information behavior and copresence on Twitch. Comput. Hum. Behav. 105, 11.
Douglas, Anneli, 2019. Mobile business travel application usage. Journal of Hospitality and Tourism Technology 10 (3), 269–285.
Eagly, Alice H., Wood, Wendy, 1991. Explaining sex differences in social behavior: a meta-analytic perspective. Pers. Soc. Psychol. Bull. 17 (3), 306–315.
Fan, X.J., Zhang, L., Guo, X., Zhao, W.Y., 2024. The impact of live-streaming interactivity on live-streaming sales mode based on game-theoretic analysis. J. Retailing Consum. Serv. 81, 10.
Fang, Jie, Wen, Zhonglin, 2022. Moderation Analyses of Two Frequently-Used Types of Categorical Variable, vol. 45, pp. 702–709.
Fang, Jie, Wen, Zhonglin, Zhang, Minqiang, 2017. Mediation effect analysis of class variables. Journal of Psychological Science 40 (2), 7.
Fang, J., Liu, H.F., Li, Y., Cai, Z., 2021. Retaining customers with in-store mobile usage experience in omni-channel retailing: the moderating effects of product information overload and alternative attractiveness. Electron. Commer. Res. Appl. 46, 13.
Feng, X.J., Rong, Y., Tian, X., Wang, M.M., Yao, Y.L., 2024. When persuasion is too persuasive: an empirical analysis of product returns in livestream e-commerce. Prod. Oper. Manag. 20.
Fu, J.R., Lu, I.W., Chen, J.H.F., Farn, C.K., 2020. Investigating consumers' online social shopping intention: an information processing perspective. Int. J. Inf. Manag. 54, 18.
Furner, C.P., Zinko, R., Zhu, Z., 2016. Electronic word-of-mouth and information overload in an experiential service industry. J Serv Theory Pract 26 (6), 788–810.
Ganguly, Boudhayan, Bhusan, Dash Satya, Dianne, Cyr, Milena, Head, 2010. The effects of website design on purchase intention in online shopping: the mediating role of trust and the moderating role of culture. Int. J. Electron. Bus. 8 (4–5), 302–330.
Gao, W., Liu, Z.P., Guo, Q.Q., Li, X., 2018. The dark side of ubiquitous connectivity in smartphone-based SNS: an integrated model from information perspective. Comput. Hum. Behav. 84, 185–193.
Gao, X.Y., Xu, X.Y., Tayyab, S.M.U., Li, Q., 2021. How the live streaming commerce viewers process the persuasive message: an ELM perspective and the moderating effect of mindfulness. Electron. Commer. Res. Appl. 49, 15.
Gao, W., Jiang, N., Gu, F., 2023. Understanding the role of streamers in livestreaming commerce: a vocal-visual perspective. J Manag Anal 10 (2), 247–269.
Geng, S., Yang, P.P., Gao, Y., Tan, Y.S., Yang, C.C., 2021. The effects of ad social and personal relevance on consumer ad engagement on social media: the moderating role of platform trust. Comput. Hum. Behav. 122, 12.
Gottschalk, F., Mimra, W., Waibel, C., 2020. Health services as credence goods: a field experiment. Econ. J. 130 (629), 1346–1383.
Gu, Yu, Cheng, Xusen, Jia, Shen, 2023. Design shopping as an experience: exploring the effect of the live-streaming shopping characteristics on consumers' participation intention and memorable experience. Inf. Manag. 60 (5), 103810.
Guo, Y.Y., Zhang, K.X., Wang, C.Y., 2022. Way to success: understanding top streamer's popularity and influence from the perspective of source characteristics. J. Retailing Consum. Serv. 64, 13.
Halilovic, S., Cicic, M., 2013. Antecedents of information systems user behaviour extended expectation-confirmation model. Behav. Inf. Technol. 32 (4), 359–370.
Harwood, T., Garry, T., 2015. An investigation into gamification as a customer engagement experience environment. J. Serv. Market. 29 (6–7), 533–546.
Hashmi, Hamna, Saman, Attiq, Farooq, Rasheed, 2019. Factors affecting online impulsive buying behavior: a stimulus organism response model approach. Market forces 14 (1).
Hausman, Angela, 2000. A multi-method investigation of consumer motivations in impulse buying behavior. J. Consum. Market. 17 (5), 403–426.
Hilvert-Bruce, Z., Neill, J.T., Sjöblom, M., Hamari, J., 2018. Social motivations of live-streaming viewer engagement on Twitch. Comput. Hum. Behav. 84, 58–67.
Hofacker, C.F., de Ruyter, K., Lurie, N.H., Manchanda, P., Donaldson, J., 2016. Gamification and mobile marketing effectiveness. J. Interact. Market. 34, 25–36.
Hoffman, D.L., Novak, T.P., 1996. Marketing in hypermedia computer-mediated environments: conceptual foundations. J. Market. 60 (3), 50–68.
Hollebeek, L.D., Macky, K., 2019. Digital content marketing's role in fostering consumer engagement, trust, and value: framework, fundamental propositions, and implications. J. Interact. Market. 45, 27–41.
Hollebeek, L.D., Glynn, M.S., Brodie, R.J., 2014. Consumer brand engagement in social media: conceptualization, scale development and validation. J. Interact. Market. 28 (2), 149–165.
Hollebeek, L.D., Keeling, D.I., de Ruyter, K., 2022. Customer engagement design in industrial innovation. Ind. Mark. Manag. 106, 83–89.
Hollebeek, L.D., Sarstedt, M., Menidjel, C., Sprott, D.E., Urbonavicius, S., 2023. Hallmarks and potential pitfalls of customer- and consumer engagement scales: a systematic review. Psychol. Market. 40 (6), 1074–1088.
Hu, Han-fen, Krishen, Anjala S., 2019. When is enough, enough? Investigating product reviews and information overload from a consumer empowerment perspective. J. Bus. Res. 100, 27–37.
Hu, Mu, Zhang, Mingli, Wang, Yu, 2017. Why do audiences choose to keep watching on live video streaming platforms? An explanation of dual identification framework. Comput. Hum. Behav. 75, 594–606.
Huang, L.T., 2016. Flow and social capital theory in online impulse buying. J. Bus. Res. 69 (6), 2277–2283.
Huang, T.L., Chung, H.F.L., 2024. Impact of delightful somatosensory augmented reality experience on online consumer stickiness intention. J. Res. Indian Med. 18 (1), 6–30.
Huang, Qianru, Omkar, Dastane, Tat-Huei, Cham, Jun-Hwa, Cheah, 2024. Is 'she' more impulsive (to pleasure) than 'him' during livestream e-commerce shopping? J. Retailing Consum. Serv. 78, 103707.
Isaac, Anne R., Marks David, F., 1994. Individual differences in mental imagery experience: developmental changes and specialization. Br. J. Psychol. 85 (4), 479–500.
Jacoby, Jacob, 1984. Perspectives on information overload. J. Consum. Res. 10 (4), 432–435.

Jia, X.Y., Wang, R.L., Lu, Y.B., Liu, J.H., Pan, Z., 2024. Investigation of continuance stream-watching intention: an empirical study. Inf. Technol. Manag. 12.
Jones, Michael A., Taylor, Valerie A., 2018. Marketer requests for positive post-purchase satisfaction evaluations: consumer depth interview findings. J. Retailing Consum. Serv. 41, 218–226.
Joo, E., Yang, J., 2023. How perceived interactivity affects consumers' shopping intentions in live stream commerce: roles of immersion, user gratification and product involvement. J. Res. Indian Med. 17 (5), 754–772.
Kang, K., Lu, J.X., Guo, L.Y., Li, W.L., 2021. The dynamic effect of interactivity on customer engagement behavior through tie strength: evidence from live streaming commerce platforms. Int. J. Inf. Manag. 56, 16.
Kim, Dae-Young, Lehto, Xinran Y., Morrison, Alastair M., 2007. Gender differences in online travel information search: implications for marketing communications on the internet. Tour. Manag. 28 (2), 423–433.
Koufaris, M., 2002. Applying the technology acceptance model and flow theory to online consumer behavior. Inf. Syst. Res. 13 (2), 205–223.
Kumar, Amresh, Bhawna, Anjaly, 2017. How to measure post-purchase customer experience in online retailing? A scale development study. Int. J. Retail Distrib. Manag. 45 (12), 1277–1297.
Lai, H.M., Hsieh, P.J., Zhang, R.C., 2019. Understanding adolescent students' use of Facebook and their subjective wellbeing: a gender-based comparison. Behav. Inf. Technol. 38 (5), 533–548.
Lee, M.T., Theokary, C., 2021. The superstar social media influencer: exploiting linguistic style and emotional contagion over content? J. Bus. Res. 132, 860–871.
Leppink, J., Paas, F., van Gog, T., van der Vleuten, C.P.M., van Merriënboer, J.J.G., 2014. Effects of pairs of problems and examples on task performance and different types of cognitive load. Learn. InStruct. 30, 32–42.
Li, C.Y., 2019. How social commerce constructs influence customers' social shopping intention? An empirical study of a social commerce website. Technol. Forecast. Soc. Change 144, 282–294.
Li, M.C., Deng, R.C., Gong, B., 2024. Research on the impact of live marketing on consumers' irrational consumption behavior under the background of the new economic era. J. Knowl. Econ. 50.
Li, Y., Peng, Y., 2021. What drives gift-giving intention in live streaming? The perspectives of emotional attachment and flow experience. Int. J. Hum. Comput. Interact. 37 (14), 1317–1329.
Liang, T.P., Lai, H.J., Ku, Y.C., 2006. Personalized content recommendation and user satisfaction: theoretical synthesis and empirical findings. J. Manag. Inf. Syst. 23 (3), 45–70.
Liao, Chechen, Hong-Nan, Lin, Meiling, Luo Margaret, Sophea, Chea, 2017. Factors influencing online shoppers' repurchase intentions: the roles of satisfaction and regret. Inf. Manag. 54 (5), 651–668.
Lin, Yan, Dai, Yao, Chen, Xingyu, 2021. Happiness begets money: emotion and engagement in live streaming. J. Mark. Res. 58 (3), 417–438.
Liu, L.L., Lee, M.K.O., Liu, R.J., Chen, J.W., 2018. Trust transfer in social media brand communities: the role of consumer engagement. Int. J. Inf. Manag. 41, 1–13.
Liu, P., Li, M.Q., Dai, D., Guo, L.Y., 2021. The effects of social commerce environmental characteristics on customers' purchase intentions: the chain mediating effect of customer-to-customer interaction and customer-perceived value. Electron. Commer. Res. Appl. 48, 14.
Liu, Zilong, Jianing, Li, Xuequn, Wang, Yiru, Guo, 2023. How search and evaluation cues influence consumers' continuous watching and purchase intentions: an investigation of live-stream shopping from an information foraging perspective. J. Bus. Res. 168, 114233.
Liu, Y.F., Zhang, M.L., Hu, M., Qiao, T., 2025. How multi-actor resources create value for live streaming platforms: the mediating role of engaged spectators. Serv. Ind. J. 45 (5–6), 526–555.
Lo, Pei-San, Dwivedi, Yogesh K., Garry, Wei-Han Tan, Keng-Boon, Ooi, Aw Eugene, Cheng-Xi, Bhimaraya, Metri, 2022. Why do consumers buy impulsively during live streaming? A deep learning-based dual-stage SEM-ANN analysis. J. Bus. Res. 147, 325–337.
Luo, Lijuan, Xu, Meiling, Zheng, Yujie, 2024a. Informative or affective? Exploring the effects of streamers' topic types on user engagement in live streaming commerce. J. Retailing Consum. Serv. 79, 103799.
Luo, X., Cheah, J.H., Hollebeek, L.D., Lim, X.J., 2024b. Boosting customers' impulsive buying tendency in live-streaming commerce: the role of customer engagement and deal proneness. J. Retailing Consum. Serv. 77, 15.
Lv, X.F., Li, J., Wang, Q.Y., 2024. The dark side of recommendation algorithms in Chinese mass short video apps: effect of perceived over-recommendation on users' cognitive dissonance and discontinuance intention. Int. J. Hum. Comput. Interact. 15.
Ma, Z.Y., Gao, Q., Chen, Y., 2023. Post-purchase trust in e-commerce: a theoretical framework and a text mining-based assessment method. Int. J. Hum. Comput. Interact. 39 (8), 1643–1661.
Malhotra Naresh, K., 1982. Information load and consumer decision making. J. Consum. Res. 8 (4), 419–430.
Malhotra Naresh, K., 1984. Reflections on the information overload paradigm in consumer decision making. J. Consum. Res. 10 (4), 436–440.
McClelland, David C., 1975. Power: the Inner Experience. Irvington, Oxford, England.
Meyers-Levy, Joan, 1988. The influence of sex roles on judgment. J. Consum. Res. 14 (4), 522–530.
Meyers-Levy, Joan, 1991. Elaborating on elaboration: the distinction between relational and item-specific elaboration. J. Consum. Res. 18 (3), 358–367.
Meyers-Levy, Joan, Durairaj, Maheswaran, 1991. Exploring differences in males' and females' processing strategies. J. Consum. Res. 18 (1), 63–70.
Mohr, Jakki J., Sohi, Ravipreet S., 1995. Communication flows in distribution channels: impact on assessments of communication quality and satisfaction. J. Retailing 71 (4), 393–415.
Nadeem, Waqar, Ming, Tan Teck, Mina, Tajvidi, Nick, Hajli, 2021. How do experiences enhance brand relationship performance and value co-creation in social commerce? The role of consumer engagement and self brand-connection. Technol. Forecast. Soc. Change 171, 120952.
Oghuma, A.P., Libaque-Saenz, C.F., Wong, S.F., Chang, Y., 2016. An expectation-confirmation model of continuance intention to use mobile instant messaging. Telematics Inf. 33 (1), 34–47.
Oliveira, T., Alhinho, M., Rita, P., Dhillon, G., 2017. Modelling and testing consumer trust dimensions in e-commerce. Comput. Hum. Behav. 71, 153–164.
Oliver, Richard L., 1980. A cognitive model of the antecedents and consequences of satisfaction decisions. J. Mark. Res. 17 (4), 460–469.
Paul, S., Nazareth, D.L., 2010. Input information complexity, perceived time pressure, and information processing in GSS-based work groups: an experimental investigation using a decision schema to alleviate information overload conditions. Decis. Support Syst. 49 (1), 31–40.
Peng, Lifang, Zhang, Na, Huang, Ling, 2024. How the source dynamism of streamers affects purchase intention in live streaming e-commerce: considering the moderating effect of Chinese consumers' gender. J. Retailing Consum. Serv. 81, 103949.
Petty, Richard, John, Cacioppo, Rachel, Goldman, 1981. Personal involvement as a determinant of argument-based persuasion. J. Pers. Soc. Psychol. 41, 847–855.
Puccinelli, Nancy M., Rajesh, Chandrashekaran, Dhruv, Grewal, Rajneesh, Suri, 2013. Are men seduced by red? The effect of red versus black prices on price perceptions. J. Retailing 89 (2), 115–125.
Quester, Pascale G., Smart, Justin, 1998. The influence of consumption situation and product involvement over consumers' use of product attribute. J. Consum. Market. 15 (3), 220–238.
Rather, R.A., 2018. Investigating the impact of customer brand identification on hospitality brand loyalty: a social identity perspective. J. Hospit. Market. Manag. 27 (5), 487–513.
Ren, Shengnan, Karimi, Sahar, Alberto, Bravo Velázquez, Cai, Jianfeng, 2023. Endorsement effectiveness of different social media influencers: the moderating effect of brand competence and warmth. J. Bus. Res. 156, 113476.
Rook, Dennis W., 1987. The buying impulse. J. Consum. Res. 14 (2), 189–199.
Sandra Lipsitz, Bem, 1981. Gender schema theory: a cognitive account of sex typing. Psychol. Rev. 88 (4), 354–364.
Sarwar, M.A., Nasir, J., Sarwar, B., Hussain, M., Abbas, A., 2024. An investigation of precursors of online impulse buying and its effects on purchase regret: role of consumer innovation. Int. J. Innovat. Sci. 16 (5), 877–894.
Scheibehenne, Benjamin, Greifeneder, Rainer, Todd Peter, xa, John Deighton, M., 2010. Can there ever Be too many options? A meta-analytic review of choice overload. J. Consum. Res. 37 (3), 409–425 served as editor, Steve Hoch served as associate editor for this article.
Sergio, Barta, Raquel, Gurrea, Carlos, Flavián, 2022. The role of flow consciousness in consumer regret. Internet Res. 32 (3), 875–896.
Sharon, Beatty, Morgan, Robert, 2012. Customer engagement: exploring customer relationships beyond purchase. J. Market. Theor. Pract. 20, 127–145.
Shen, H.W., Zhao, C., Fan, D.X.F., Buhalis, D., 2022. The effect of hotel livestreaming on viewers? purchase intention: exploring the role of parasocial interaction and emotional engagement. Int. J. Hospit. Manag. 107, 13.
Smith, T.E., Leaper, C., 2006. Self-perceived gender typicality and the peer context during adolescence. J. Res. Adolesc. 16 (1), 91–103.
Sun, Y., Shao, X., Li, X.T., Guo, Y., Nie, K., 2019. How live streaming influences purchase intentions in social commerce: an IT affordance perspective. Electron. Commer. Res. Appl. 37, 12.
Swar, B., Hameed, T., Reychav, I., 2017. Information overload, psychological ill-being, and behavioral intention to continue online healthcare information search. Comput. Hum. Behav. 70, 416–425.
Thakur, R., 2019. The moderating role of customer engagement experiences in customer satisfaction-loyalty relationship. Eur. J. Market. 53 (7), 1278–1310.
Tian, Y., Frank, B., 2024. Optimizing live streaming features to enhance customer immersion and engagement: a comparative study of live streaming genres in China. J. Retailing Consum. Serv. 81, 16.
Tsichla, Eirini, Hatzithomas, Leonidas, Christina, Boutsouki, 2016. Gender differences in the interpretation of web atmospherics: a selectivity hypothesis approach. J. Market. Commun. 22 (6), 563–586.
van Doorn, J., Lemon, K.N., Mittal, V., Nass, S., Pick, D., Pirner, P., Verhoef, P.C., 2010. Customer engagement behavior: theoretical foundations and research directions. J. Serv. Res. 13 (3), 253–266.
Viglia, G., Pera, R., Dyussembayeva, S., Mifsud, M., Hollebeek, L.D., 2023. Engagement and value cocreation within a multi-stakeholder service ecosystem. J. Bus. Res. 157, 9.
Vivek, Shiri D., Beatty Sharon, E., Customer Engagement, Morgan Robert M., 2012. Exploring customer relationships beyond purchase. J. Market. Theor. Pract. 20 (2), 122–146.
Wang, W.T., Wang, Y.S., Liu, E.R., 2016. The stickiness intention of group-buying websites: the integration of the commitment-trust theory and e-commerce success model. Inf. Manag. 53 (5), 625–642.
Wang, Y.Q., Liu, G.H., Zhu, L.J., Nie, K., Qiong, W., 2024. Matching is believing: the effect of congruency on purchase intention in live streaming shopping context. Curr. Psychol. 15.
Wongkitrungrueng, A., Assarut, N., 2020. The role of live streaming in building consumer trust and engagement with social commerce sellers. J. Bus. Res. 117, 543–556.

Wu, I.L., Chen, K.W., Chiu, M.L., 2016. Defining key drivers of online impulse purchasing: a perspective of both impulse shoppers and system users. Int. J. Inf. Manag. 36 (3), 284–296.
Wu, I.L., Chiu, M.L., Chen, K.W., 2020. Defining the determinants of online impulse buying through a shopping process of integrating perceived risk, expectation-confirmation model, and flow theory issues. Int. J. Inf. Manag. 52, 12.
Xi, Nannan, Juho, Hamari, 2020. Does gamification affect brand engagement and equity? A study in online brand communities. J. Bus. Res. 109, 449–460.
Xiang, L., Zheng, X.B., Lee, M.K.O., Zhao, D.T., 2016. Exploring consumers' impulse buying behavior on social commerce platform: the role of parasocial interaction. Int. J. Inf. Manag. 36 (3), 333–347.
Xin, Meiling, Liu, Wei, Ling, Jian, 2024. Live streaming product display or social interaction: how do they influence consumer intention and behavior? A heuristic-systematic perspective. Electron. Commer. Res. Appl. 67, 101437.
Xiong, Jie, Wang, Yawei, Zheng, Li, 2024. Understanding the relationship between IT affordance and consumers' purchase intention in E-commerce live streaming: the moderating effect of gender. Int. J. Hum. Comput. Interact. 40 (20), 6303–6313.
Xu, Wei, Zhang, Xiaohan, Chen, Runyu, Yang, Zhan, 2023. How do you say it matters? A multimodal analytics framework for product return prediction in live streaming e-commerce. Decis. Support Syst. 172, 113984.
Xu, Xu, Luo, Chuan, Luo, Xin, Wang, Zeen, 2024. Examining how emotions affect online audience retention: empirical evidence from livestreaming electronic commerce platforms. Inf. Manag. 61 (7), 104031.
Yan, Ying, Chen, Hongquan, Shao, Bingjia, Yuanyang, Lei, 2023. How IT affordances influence customer engagement in live streaming commerce? A dual-stage analysis of PLS-SEM and fsQCA. J. Retailing Consum. Serv. 74, 103390.
Yang, T.T., Yang, F., Men, J., 2023. Understanding consumers' continuance intention toward recommendation vlogs: an exploration based on the dual-congruity theory and expectation-confirmation theory. Electron. Commer. Res. Appl. 59, 12.
Yen, C.S., Chen, G.L., Kang, C.C., Wang, Y.H., Yang, S.C., 2024. Investigating factors that influence purchase intentions in live-streaming contexts through the elaboration likelihood model: the perspectives of para-social interaction and information quality. Rev Socionetwork Strateg 18 (2), 387–406.
Yim, M.Y.C., Chu, S.C., Sauer, P.L., 2017. Is augmented reality technology an effective tool for E-commerce? An interactivity and vividness perspective. J. Interact. Market. 39, 89–103.
Zaichkowsky, Judith Lynne, 1985. Measuring the involvement construct. J. Consum. Res. 12 (3), 341–352.
Zeng, Qingfeng, Qian, Guo, Zhuang, Wei, Zhang, Yu, Fan, Weiguo, 2023. Do real-time reviews matter? Examining how bullet screen influences consumers' purchase intention in live streaming commerce. Inf. Syst. Front. 25 (5), 2051–2067.
Zhai, Mengfan, Chen, Yuan, 2023. How do relational bonds affect user engagement in e-commerce livestreaming? The mediating role of trust. J. Retailing Consum. Serv. 71, 103239.
Zhang, Q.Y., Ahmad, W., 2024. Online impulse purchase in social commerce: roles of social capital and information overload. Int. J. Hum. Comput. Interact. 40 (16), 4412–4429.
Zhang, N., Ruan, C.H., 2024. Danmaku consistency reduces consumer purchases during live streaming: a dual-process model. Psychol. Market. 41 (11), 2591–2607.
Zhang, Y.F., Xu, Q., 2024. Consumer engagement in live streaming commerce: value co-creation and incentive mechanisms. J. Retailing Consum. Serv. 81, 14.
Zhang, X.P., Zhang, S.H., 2024. Investigating impulse purchases in live streaming e-commerce: a perspective of match-ups. Technol. Forecast. Soc. Change 205, 9.
Zhang, Han, Yang, Yang, Jichang, Zhao, 2023. Does game-irrelevant chatting stimulate high-value gifting in live streaming? A session-level perspective. Comput. Hum. Behav. 138, 107467.
Zhao, L., Zhang, M.L., Ming, Y.X., Niu, T., Wang, Y., 2023. The effect of image richness on customer engagement: evidence from Sina Weibo. J. Bus. Res. 154, 14.
Zheng, R., Li, Z., Na, S., 2022. How customer engagement in the live-streaming affects purchase intention and customer acquisition, E-tailer?s perspective. J. Retailing Consum. Serv. 68, 10.
Zheng, S.Y., Chen, J.D., Liao, J.Y., Hu, H.L., 2023. What motivates users? viewing and purchasing behavior motivations in live streaming: a stream-streamer-viewer perspective. J. Retailing Consum. Serv. 72, 10.